

Convergence is not correctness: context-dependent performance of enhanced-sampling methods across biological complexity

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Enhanced-sampling molecular dynamics is indispensable for studying biomolecular processes beyond the reach of conventional simulation, yet its reliability in realistic biological settings remains poorly characterized. Here, we compare four widely used approaches—replica-exchange umbrella sampling (REUS), well-tempered metadynamics (WT-MtD), well-tempered metadynamics–extended adaptive-biasing force (WTM-eABF), and on-the-fly probability enhanced sampling (OPES)—across five experimentally grounded processes: ligand binding, conformational change, kinase activation, membrane permeation, and ion conduction. Three findings emerge. Method performance does not transfer across biological contexts; each approach exhibits characteristic failure modes invisible in low-dimensional model systems but consequential in realistic landscapes. Self-consistency metrics, the field’s standard convergence diagnostic, are insufficient: simulations can satisfy convergence criteria while yielding quantitatively incorrect free-energy profiles. Cross-method agreement against an independent reference is therefore necessary. These results re-frame method selection as a context-dependent decision and argue that cross-method validation should become a community standard.

Many biological processes of central importance—signal transduction, molecular recognition, membrane remodeling, or protein folding—take place on timescales that vastly exceed the millisecond regime¹. Capturing such events at atomic resolution would provide a dynamic view of molecular function, clarifying how large-scale domain rearrangements, often coupled through allosteric networks, emerge from local motions. Atomistic molecular dynamics (MD) simulations offer a direct window to probe into these processes, with the potential to establish the precise sequence of conformational transitions induced

by external stimuli. Yet, despite remarkable progress in computational power and algorithms, the gap between accessible and biologically relevant timescales remains substantial². Even the most advanced MD engines, reaching hundreds of nanoseconds per day on supercomputers or several microseconds on specialized hardware^{3,4}, fall far short of the milliseconds-to-seconds range characteristic of many rare events⁵. Thus, understanding slow molecular phenomena requires more than raw computational speed—it demands conceptual and methodological innovation.

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In principle, MD simulations explore the phase space of a system in an ergodic manner. In practice, however, they often fail to do so because of limited sampling efficiency. When molecular motion is hindered by high free-energy barriers, the system becomes kinetically trapped in metastable states, and transitions between them appear as rare, statistically elusive events⁶. This apparent non-ergodicity is not a flaw of the theory, but a symptom of insufficient sampling^{7,8}. Enhanced-sampling methods aim precisely to overcome this limitation by selectively accelerating the exploration of relevant regions of configurational space. They act on a subset of slow collective degrees of freedom (DOFs), guiding the system along physically meaningful transition pathways, and enabling the reconstruction of free-energy landscapes (FELs) that would otherwise remain inaccessible^{9,10}.

A central concept shared by most enhanced-sampling techniques is that of a progress variable or reaction coordinate (RC) model, a reduced descriptor that tracks the evolution of the system between metastable states. Assuming a separation of timescales, the RC is typically expressed in terms of a few collective variables (CVs), chosen to capture the essential slow modes driving the transition while averaging out the fast ones¹¹. This assumption, while powerful, entails a delicate trade-off. An overly simplified representation of the relevant DOFs often leads to a distorted view of the reaction mechanism and sluggish convergence of the simulation. Although the free-energy difference between states is path-independent, the efficiency and reliability of its computation depend critically on how the CVs are defined. Their selection cannot rely solely on intuition or visual inspection. As a result, a broad spectrum of techniques—including dimensionality reduction, spectral methods, and machine-learning-based approaches—has been developed to infer optimal CVs directly from simulation data^{12–14}.

Accelerating the sampling of rare events typically involves encouraging exploration along a surrogate RC, $\xi(\mathbf{x})$, function of Cartesian coordinates $\mathbf{x}$, through the application of an external biasing potential. This bias lowers the effective free-energy barriers, facilitating transitions and improving statistical convergence. A diverse array of methods has been proposed for this purpose, each embodying distinct philosophies—adaptive-biasing force¹⁵, metadynamics⁶, umbrella sampling¹⁶, among many others. The growing diversity of algorithms has made it increasingly challenging to identify the most appropriate strategy for a given problem. For this reason, it is now essential to develop unified frameworks that clarify the conceptual connections among existing approaches, helping practitioners navigate this ever expanding landscape, and select methods best suited to their scientific goals. Here, we focus on four members of the enhanced-sampling family of algorithms that have become central to contemporary practice, namely replica-exchange umbrella sampling (REUS)¹⁷, well-tempered metadynamics (WT-MtD)¹⁸, well-tempered metadynamics-extended adaptive-biasing force (WTM-eABF)^{19,20}, and on-the-fly probability enhanced sampling (OPES)²¹. Together, these algorithms span the dominant conceptual strategies—adaptive-biasing, stratification, and hybrid extended-system formulations, used to overcome rare-event sampling limitations in biomolecular simulations.

Despite their widespread adoption, these methods have been validated almost exclusively on low-dimensional model systems—most commonly the *N*-acetyl-*N'*-methylalaninamide (NANMA), often referred to as the alanine dipeptide—that bear little resemblance to the biological complexity practitioners ultimately aim to describe. Whether four leading approaches recover the same thermodynamic signatures when applied to the same realistic biomolecular systems, and whether the standard convergence diagnostics used to assess them are sufficient to detect when they do not, has not been rigorously tested in realistic biological settings. This gap carries serious practical risk: We show that methods can appear converged by standard criteria while producing qualitatively incorrect free-energy profiles, meaning that

the reliability of a substantial body of published free-energy calculations rests on assumptions about transferability from toy models to molecular reality that have never been validated.

In the present work, we move decisively beyond these paradigmatic cases and assess the performance of all four enhanced-sampling algorithms across a series of five experimentally grounded biomolecular processes and mechanistic classes—protein-ligand binding (Abl-SH3:p41), coupled binding and conformational transition (ribose-binding protein binding ribose), conformational rearrangement (BCR-Abl kinase DFG-in/out activation), membrane permeation (2', 3'-dideoxyadenosine crossing a lipid bilayer), and channel permeation by an ion (nAChR with Na⁺) (see Fig. 1). Performance is evaluated using both a self-consistency metric based on the time evolution of the free energy RMSD, and cross-method comparison against both a marginalized consensus FEL and independent, long multiple-walker (MW) WTM-eABF reference trajectories. Three findings emerge with implications for the field as a whole, namely (i) method performance does not transfer across biological complexity, (ii) standard convergence diagnostics can fail silently, and (iii) cross-method validation is not optional—it is necessary. Together, these results argue for a concrete change in how enhanced-sampling free-energy calculations are performed, validated, and reported.

Results

The FELs reported in the following sections were generated using four distinct enhanced-sampling methodologies. Each simulation was performed independently and blindly, without a priori knowledge of the FELs obtained by the different approaches. This strategy eliminates bias in simulation-stopping criteria and ensures that the observed level of agreement among methods arises from genuine convergence rather than targeted tuning. One should note that the parameters of the enhanced-sampling methods were chosen according to established best practices such that further tuning would be expected to yield only marginal improvements. Convergence of each simulation was first monitored using a self-consistency criterion based on the root-mean-square deviation (RMSD) of the FEL evolving over time with respect to its final estimate. A simulation was stopped only when this RMSD plateaued at a value $\leq k_B T$, indicating that continued sampling would not result in significant improvement to the characteristics of the FEL. The convergence times are reported in Table 1; For REUS, this is the aggregate sampling time across all umbrella windows, but for WT-MtD, WTM-eABF, and OPES, it is the total single-replica (or, where stratification was used, multi-window) simulation time. Simulations failing to satisfy this criterion are flagged as not converged (NC) in Table 1. To then contextualize individual FELs, the relative performance of the different enhanced-sampling schemes was evaluated against two complementary references. The first is a marginalized average constructed from all final FELs for a given system, which serves as an internal consensus of the underlying FEL. This metric is used to reduce the influence of method-specific artifacts and highlights commonalities across enhanced-sampling schemes. The second reference consists of long, multi- μ s, multiple-walker^{22,23} (MW) WTM-eABF trajectories. Although the adoption of this particular method as a reference may appear biased, the extended timescale should render the bias increasingly negligible. The subsequent sections, therefore, do not merely report individual FELs; instead, they quantify the extent to which different methods recover the same thermodynamic signatures and reveal which strategies are most suited to specific classes of biomolecular processes.

Protein-ligand binding Abl-SH3:p41

The potential of mean force (PMF) for the dissociation of the p41 peptide from the Abl-SH3 domain along the distance, r (Fig. 1a), separating the corresponding centers of mass is shown for all four enhanced-sampling methods in Fig. 2. Consistent with previous work²⁴,

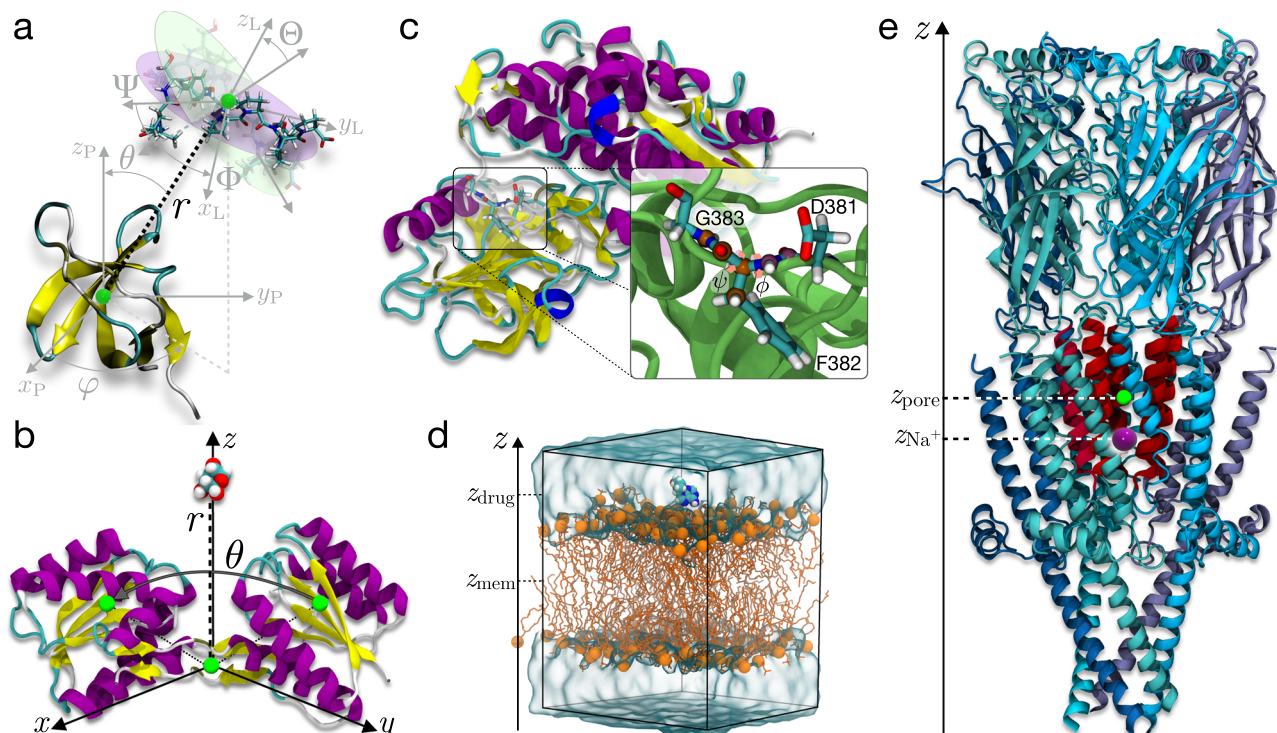


Fig. 1 | Molecular processes studied in this work and the CVs employed to characterize and enhance sampling of each process. Protein-ligand binding: Abl-SH3 domain binding proline-rich peptide p41 (a). Coupled protein-ligand binding/conformational transition: Ribose binding to the ribose-binding protein (b).

Conformational transition: BCR-Abl1 kinase activation, DFG in/out flip (c). Membrane permeation: 2', 3'-dideoxyadenosine permeating a POPC bilayer (d). Ion permeation: Sodium permeating nAChR (e).

Table 1 | Cumulative simulation time (in μs) at which the self-convergence criterion ($\text{RMSD}_{\text{self}} \leq k_B T$) was first consistently satisfied using replica-exchange umbrella sampling (REUS), well-tempered metadynamics (WT-MtD), well-tempered metadynamics-extended adaptive-biasing force (WTM-eABF), and on-the-fly probability enhanced sampling (OPES), applied to the five biomolecular processes investigated in this work

Biomolecular process	Enhanced-sampling scheme			
	REUS	WT-MtD	WTM-eABF	OPES ^{a,b}
protein-ligand binding	0.2	0.2	0.1	0.3 (18)
Abl-SH3:p41 complex				0.3 (20)
				0.8 (22.5)
				0.6 (25)
coupled protein-ligand binding and conformational transition	1.5	2.2	1.7	2.1 (10)
ribose-binding protein (RBP):ribose				NC (20)
				NC (30)
conformational transition	1.5	NC	1.0	1.4 (7)
BCR-Abl1 kinase activation, DFG in/out flip				1.4 (10)
membrane permeation	0.3	2.8	1.7	2.1 (10)
2', 3'-dideoxyadenosine permeating a POPC bilayer				1.8 (15)
				1.4 (20)
channel permeation	0.1	1.7	0.7	1.4 (5)
sodium permeating nAChR				1.9 (10)

^a Values in parentheses indicate the barrier parameter (in kcal/mol).

^b NC: Not converged within the allocated simulation time limit.

all PMFs exhibit a well-defined bound-state basin located near $r = 12 \text{ \AA}$, separated from the ligand-free regime at $r \geq 17 \text{ \AA}$ by a free-energy difference of 14–17 kcal/mol. Despite discrepancies in the sampling strategies and total simulation times, the resulting PMFs overall agree. The REUS PMF (Fig. 2a) exhibits a bound-state minimum at $r = 12.0 \text{ \AA}$ with a well depth of ca. 16.0 kcal/mol, and reaches self-convergence after ca.

0.2 μs , as indicated by the RMSD relative to its final PMF falling below $k_B T$ (Figure S1). Agreement with the marginalized average (Fig. 2e) is achieved on the same timescale (Figure S2). The WT-MtD PMF (Fig. 2b) shows a nearly identical bound-state minimum and well depth, reaching self-convergence after ca. 0.2 μs (Figure S1), while agreement with the marginalized average occurs closer to 0.3 μs (Figure S2),

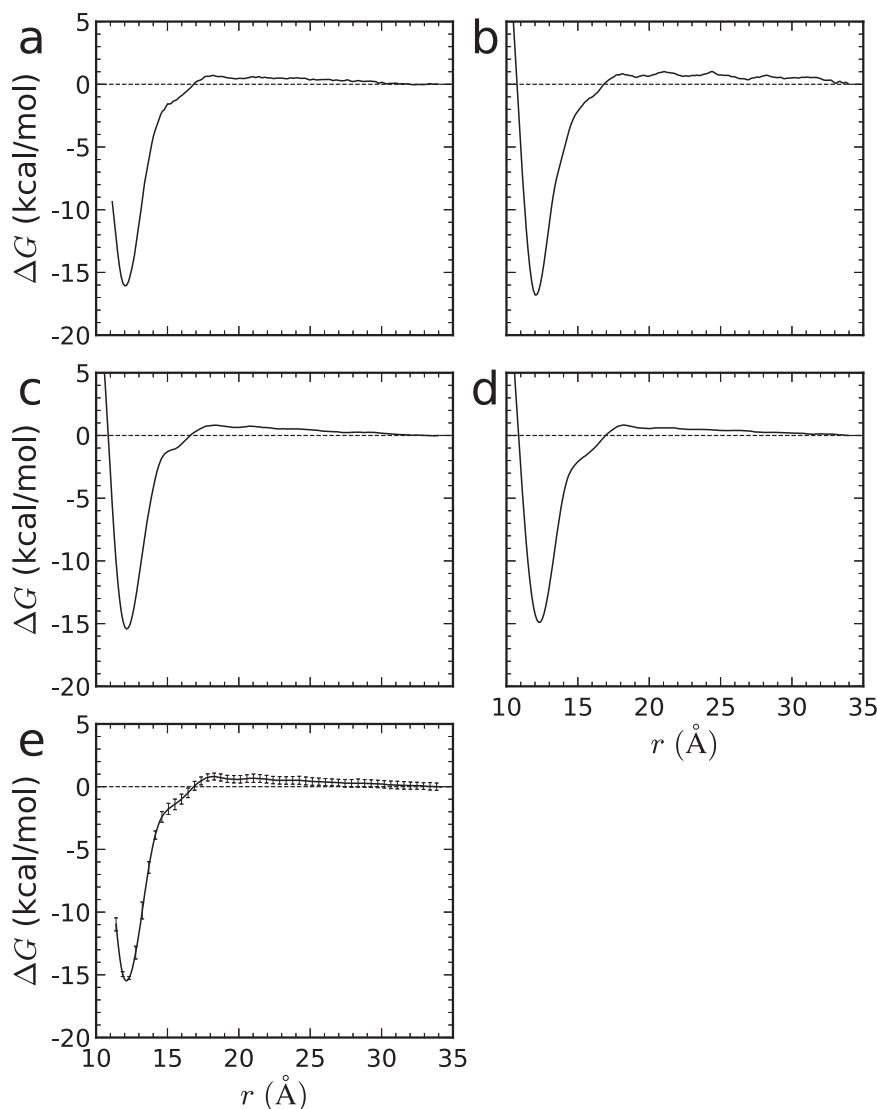


Fig. 2 | Protein-ligand binding: Abl-SH3 domain binding proline-rich peptide p41 (Fig. 1a). PMF determined using REUS (a), WT-MtD (b), WTM-eABF (c), OPES (d) and (e) average potential of mean force computed from the marginal probabilities with error bars representing one standard deviation from the mean, along

the distance, r , separating the center of mass of the ligand from that of the protein, restraining the conformation of the ligand, its relative position, as described by angles θ and ϕ , and its relative orientation, as described by Euler angles Θ , Φ and Ψ , relative to the native bound state. Source data are provided as a Source Data file.

reflecting modest residual fluctuations in the unbound regime. The WTM-eABF PMF (Fig. 2c) reproduces the same bound-state minimum and free-energy difference, and attains self-convergence after ca. 0.1 μ s (Figure S1). Notably, agreement with the marginalized average is achieved at earlier times, well before self-convergence is reached (Figure S2). Last, the OPES-derived PMF (Fig. 2d), obtained from a 1.2 μ s simulation using a barrier setting of 20 kcal/mol, exhibits the same bound-state basin near $r = 12$ Å, with a well depth of ca. 15 kcal/mol, reaching self-convergence by ca. 0.3 μ s (Figure S1) and agreement with the marginalized average by ca. 0.2 μ s (Figure S2). Although all OPES barrier settings tested here satisfy the self-convergence criterion (Figure S1), only the 20- and 22.5-kcal/mol settings yield PMFs consistent with the other methods, whereas barrier values of 18 and 25 kcal/mol produce discrepant profiles relative to the marginalized average (Figure S2), emphasizing the sensitivity of OPES to barrier selection for this system.

The marginalized average PMF for the dissociation of p41 from the Abl-SH3 domain (Fig. 2e) exhibits a bound minimum at $r = 12$ Å and a well depth of ca. 15.5 kcal/mol with minimal errors overall. The long MW WTM-eABF reference (Figure S11a) reproduces this same

minimum and reveals a well depth within ca. 1.0 kcal/mol, in close agreement with the marginalized average, and an RMSD between the two of just 0.4 kcal/mol. More broadly, all four methods yield mutually consistent PMFs for this system, with self-convergence times ranging from 0.1 μ s (WTM-eABF) to 0.3 μ s (OPES), with REUS and WT-MtD self-converging by 0.2 μ s, and agreement with the marginalized average achieved on comparable timescales.

Coupled protein-ligand binding and conformational transition of the ribose-binding protein

The two-dimensional FELs for ribose binding and conformational rearrangement in the ribose-binding protein (RBP) along the hinge angle, θ , and the distance to the ribose-binding pocket, r (Fig. 1b), are shown for all four enhanced-sampling methods in Fig. 3. Consistent with experimental observations²⁵ and prior simulations^{26,27}, three of the four methods (WT-MtD, WTM-eABF, and OPES) identify three dominant basins: A holo-closed (HC) ribose-bound state at small r and hinge angles near the native closed conformation (centered at $\theta = 111^\circ$ and $r = 10$ Å), a holo-open (HO) state ($\theta = 130^\circ$ and $r = 12$ Å), and an apo-open (AO) ligand-free state at large separations ($r = 40$ Å) with the

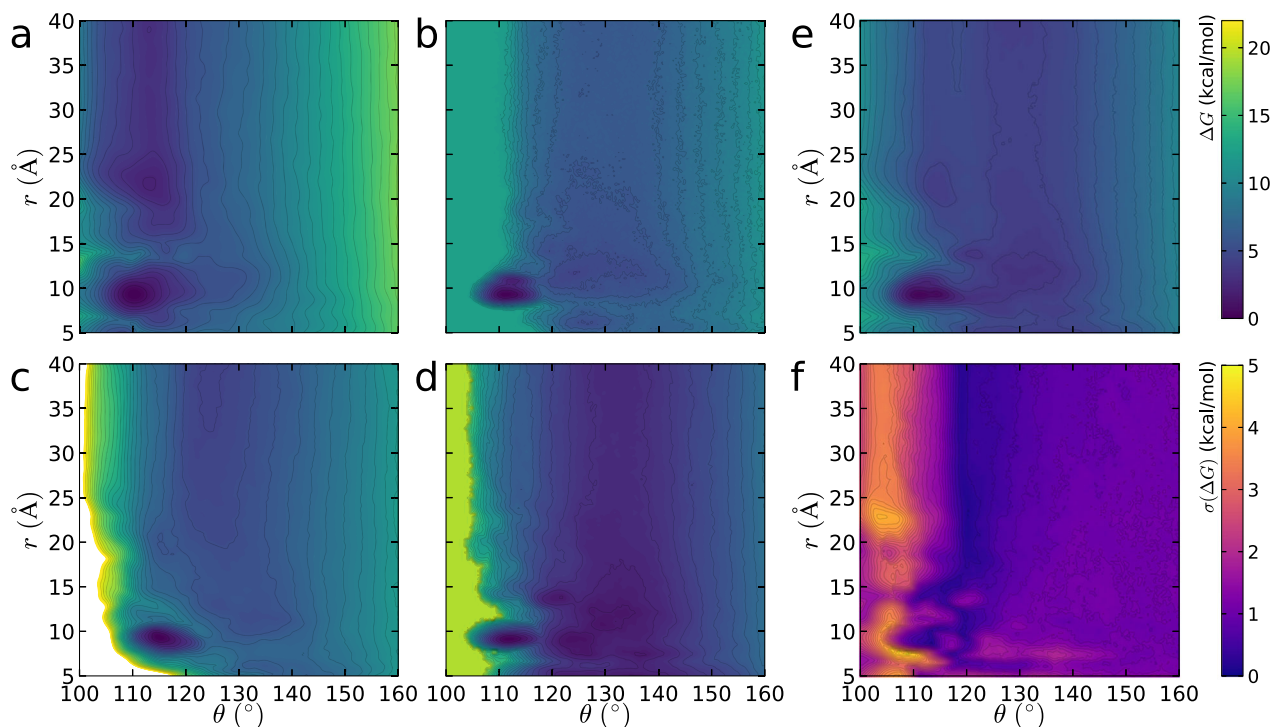


Fig. 3 | Coupled protein-ligand binding/conformational transition: Ribose binding the ribose-binding protein. FEL mapped along distance r and valence angles θ of Fig. 1b, using REUS (a), WT-MtD (b), WTM-eABF (c), and OPES (d). The average potential of mean force computed from the marginal probabilities (e) and

associated error (f). Free-energy values are colored using the perceptually uniform viridis color map; standard deviation values are colored using plasma. Source data are provided as a Source Data file.

hinge in an open conformation ($\theta = 130^\circ$). REUS recovers the HC and HO basins but, at large r , the low-free-energy corridor is shifted toward a closed hinge angle ($\theta = 114^\circ$), yielding an apo-closed (AC) state rather than the AO state found by the other three methods. As discussed below, this systematic shift can be ascribed to the seeding of the REUS windows. Free-energy differences between relevant basins were determined by integration over the corresponding regions of the FEL according to

$$\Delta G_{S \rightarrow S'} = -\frac{1}{\beta} \ln \frac{\int_S dr d\theta P(r, \theta)}{\int_{S'} dr d\theta P(r, \theta)}, \quad (1)$$

where $P(r, \theta)$ denotes the probability distribution of finding the system in a conformation described by r and θ , and S and S' define domains of interest on the FEL. The REUS FEL (Fig. 3a), obtained from ca. 1.7 μ s of sampling, reaches self-convergence after approximately 1.5 μ s (Figure S3), but maintains an RMSD of nearly 3.5 kcal/mol relative to the marginalized average (Fig. 3e) throughout the simulation (Figure S4), reflecting a systematic shift of the low-free-energy corridor toward $\theta = 114^\circ$ (AC state) rather than $\theta = 130^\circ$ (AO state), as observed with the other methods. Integration of the FEL yields $\Delta G_{\text{HC} \rightarrow \text{HO}} = 4.3$ kcal/mol, and a $\Delta G_{\text{HO} \rightarrow \text{AO}} = -1.9$ kcal/mol for the AO state relative to the HO state. The WT-MtD FEL (Fig. 3b), generated over 3.0 μ s and self-converged after ca. 2.2 μ s (Figure S3), shows the final FEL with the lowest RMSD relative to the marginalized average (ca. 1.5 kcal/mol near the end of the simulation; Figure S4). The FEL resolves the HC state ($r = 10$ Å and $\theta = 111^\circ$), a HO state ($r = 12$ Å and $\theta = 130^\circ$), and an AO state ($r = 40$ Å and $\theta = 130^\circ$). The corresponding $\Delta G_{\text{HC} \rightarrow \text{HO}} = 3.2$ kcal/mol and $\Delta G_{\text{HO} \rightarrow \text{AO}} = 1.2$ kcal/mol between the HO and AO states, yielding an overall free-energy difference of 4.4 kcal/mol between the HC and AO states. The WTM-eABF FEL (Fig. 3c), obtained from 2.5 μ s of sampling and self-converged after ca. 1.7 μ s (Figure S3), reproduces the same conformational states, with the

HC minimum slightly shifted to $r = 10$ Å and $\theta = 115^\circ$. Integration yields a $\Delta G_{\text{HC} \rightarrow \text{HO}} = 3.5$ kcal/mol between the HC and HO states, and a relative stabilization of the AO state at $\Delta G_{\text{HO} \rightarrow \text{AO}} = -1.0$ kcal/mol with respect to the HO state, resulting in a net free-energy difference of 2.5 kcal/mol between the HC and AO states. The RMSD relative to the marginalized average remains high, at ca. 4.0 kcal/mol (Figure S4), reflecting systematic deviations in the high-energy regions of the FEL. Finally, the OPES-derived FEL (Fig. 3d), obtained from 2.9 μ s of sampling and self-converged after ca. 2.1 μ s (Figure S3), captures the same dominant basins but exhibits persistent RMSD oscillations of ca. 3.5 kcal/mol relative to the marginalized average (Figure S4), once again indicating the sensitivity to the OPES barrier parameter. For the optimal barrier setting of 10 kcal/mol, integration yields a $\Delta G_{\text{HC} \rightarrow \text{HO}} = -0.3$ kcal/mol, and a $\Delta G_{\text{HO} \rightarrow \text{AO}} = 1.5$ kcal/mol between the HO and AO states. Larger barrier values (20 and 30 kcal/mol) fail to satisfy the RMSD self-convergence criterion (remaining above 3.0 kcal/mol for most of the simulation before sharply falling), after an equivalent 2.9 μ s of sampling and only the 10 kcal/mol setting yields an FEL consistent with the other methods.

The marginalized average FEL for the ribose binding and conformational rearrangement in RBP (Fig. 3e) resolves three basins corresponding to the HC, HO and AO states with the majority of error (Fig. 3f) corresponding to the high-energy regions. The long MW WTM-eABF reference (Figure S11b) recovers the same basin topography, in close agreement with the marginalized average, with an overall RMSD of ca. 5.0 kcal/mol arising from higher energy regions. However, the free energy ordering of these states is preserved.

Conformational transition during BCR–Abl1 kinase DFG-in/DFG-out activation

The two-dimensional FELs associated with the DFG-in/DFG-out conformational transition along dihedrals ψ and ϕ (Fig. 1c) in apo BCR–Abl1 kinase are shown for all enhanced-sampling methods in Fig. 4. Consistent with previous work²⁸, all methods identify two

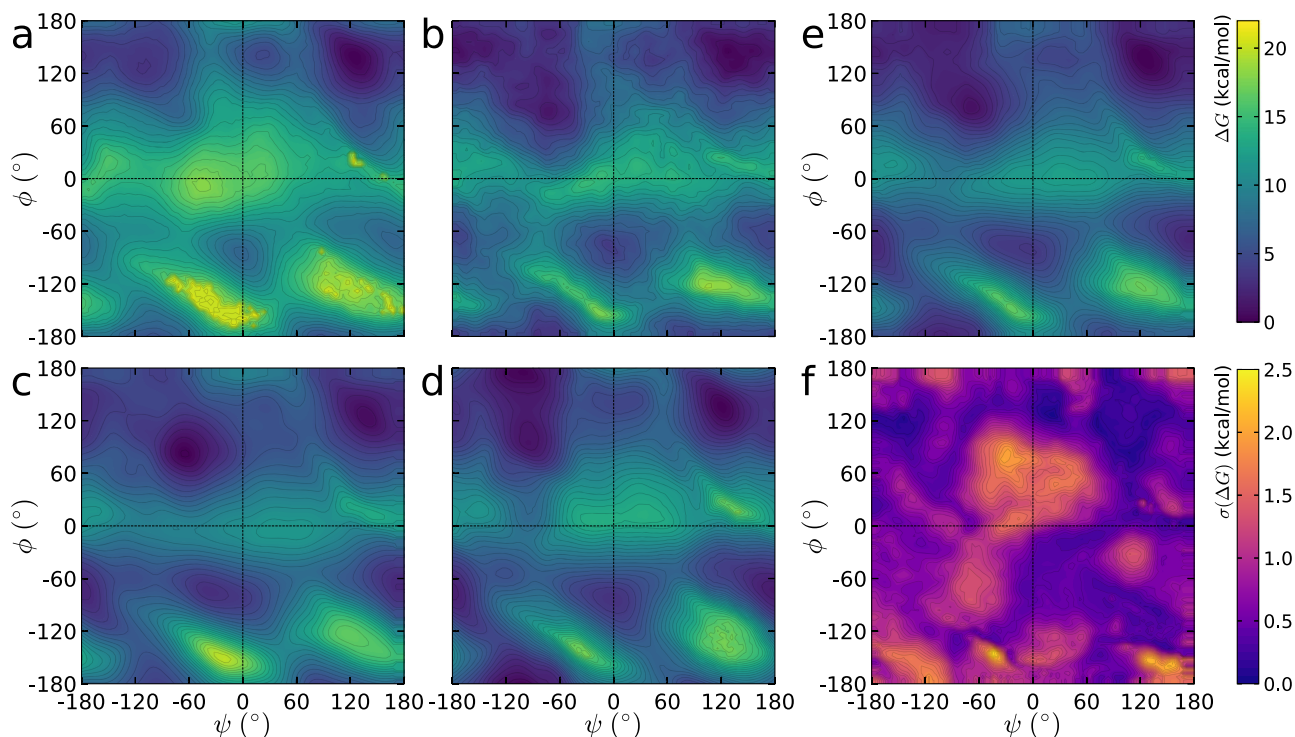


Fig. 4 | Conformational transition: BCR-AblI kinase activation, DFG in/out flip.

Free-energy landscape mapped along the ψ and ϕ dihedral angles of Fig. 1c, using REUS (a), WT-MtD (b), WTM-eABF (c), and OPES (d). The average potential of mean force computed from the marginal probabilities (e) and associated error (f). Free-

energy values are colored using the perceptually uniform viridis color map; standard deviation values are colored using plasma. Source data are provided as a Source Data file.

dominant basins corresponding to the DFG-in state, located at $\psi = -100^\circ$ and $\phi = 120^\circ$, and the DFG-out state at $\psi = +120^\circ$ and $\phi = +135^\circ$. The transition between these states is driven primarily by a 130° rotation in ψ , while ϕ undergoes comparatively smaller changes ($< 35^\circ$), confirming that the DFG flip is dominated by motion along the ψ coordinate. Free-energy differences between the two basins were quantified from each FEL using, mutatis mutandis, (1). The REUS-derived FEL (Fig. 4a), obtained from a 4.9- μ s simulation, reaches self-convergence after 1.5 μ s (Figure S5), while the RMSD relative to the marginalized average (Fig. 4e) plateaus at 2.0 kcal/mol throughout the simulation (Figure S6), chiefly due to discrepancies in high-energy regions at $\psi = -30^\circ$ and $\phi = -150^\circ$, and at $\psi = -150^\circ$ and $\phi = -120^\circ$. Integration of the REUS FEL yields a free-energy difference of 3.0 kcal/mol in favor of the DFG-out state, slightly deviating from previously reported near-isoenergetic values (e.g., $\Delta G_{\text{DFG-in} \rightarrow \text{DFG-out}} \approx 0$ kcal/mol), likely reflecting bias introduced by the initial window seeding, while accurately reproducing basin locations and the dominant ψ -driven transition pathway. The WT-MtD FEL (Fig. 4b), generated from a 2.0- μ s simulation, qualitatively resolves the same DFG-in and DFG-out basins but does not achieve full convergence, with the RMSD relative to both the final FEL and the marginalized average remaining above 2.0 kcal/mol for most of the simulation before decreasing to 1.0 kcal/mol at 2.0 μ s (Figures S5 and S6). Nevertheless, integration of the WT-MtD FEL yields $\Delta G_{\text{DFG-in} \rightarrow \text{DFG-out}} = 1.2$ kcal/mol, and secondary minima at $\psi = 0^\circ$ and $\phi = -80^\circ$ and at $\psi = -180^\circ$ and $\phi = -70^\circ$ appear ca. 3.0 kcal/mol above the dominant basins. The WTM-eABF FEL (Fig. 4c), obtained from a 1.5- μ s simulation and self-converged after 1.0 μ s (Figure S5), resolves the DFG-in basin at $\psi = -60^\circ$ and $\phi = +80^\circ$ and the DFG-out basin at $\psi = +140^\circ$ and $\phi = +120^\circ$, with additional higher-lying minima at $\psi = -40^\circ$ and $\phi = -70^\circ$, and at $\psi = -180^\circ$ and $\phi = -70^\circ$, which are 2.0 kcal/mol above the dominant states. Agreement with the marginalized average is achieved rapidly, with RMSD values fluctuating between 1.0 and 1.5 kcal/mol after 0.4 μ s (Figure S6). Integration of the

WTM-eABF FEL yields $\Delta G_{\text{DFG-in} \rightarrow \text{DFG-out}} = 0.4$ kcal/mol in favor of the DFG-in state, consistent with a near-zero free-energy difference, although the free-energy barrier separating the two states is nearly twice as large. Last, the OPES-derived FEL (Fig. 4d), generated over a 2.0- μ s simulation, reaches self-convergence after 1.4 μ s (Figure S5), while agreement with the marginalized average is achieved earlier at 0.8 μ s, with an RMSD of 1.2 kcal/mol (Figure S6). Using a barrier setting of 10 kcal/mol, OPES correctly resolves the DFG-in basin at $\psi = -95^\circ$ and $\phi = 95^\circ$, and the DFG-out basin at $\psi = +125^\circ$ and $\phi = +130^\circ$, yielding $\Delta G_{\text{DFG-in} \rightarrow \text{DFG-out}} = 0.1$ kcal/mol in favor of the DFG-out state. Secondary minima at $\psi = 0^\circ$ and $\phi = -80^\circ$ and at $\psi = -180^\circ$ and $\phi = -70^\circ$ appear at ca. 2.0 kcal/mol higher than the dominant basins, similar to those observed with WTM-eABF. Reduction of the OPES barrier to 7 kcal/mol produces incorrect relative basin populations despite apparently satisfactory self-convergence (Figure S5), underscoring the importance of appropriate barrier selection.

The marginalized average FEL for the DFG transition in apo BCR-AblI kinase (Fig. 4e) reveal two primary basins corresponding to the DFG-in and DFG-out conformations with a near isoenergetic balance ($\Delta G_{\text{DFG-in} \rightarrow \text{DFG-out}} = 0.5$ kcal/mol) with low error near the basins (Fig. 4f). The long MW WTM-eABF reference (Figure S11c) quantitatively reproduces the same basin locations and free energy difference ($\Delta G_{\text{DFG-in} \rightarrow \text{DFG-out}} = 0.4$ kcal/mol), with an RMSD between the two FELs of 1.2 kcal/mol.

Membrane permeation by 2', 3'-dideoxyadenosine

The membrane permeation PMFs for DDA across a POPC bilayer characterized along the z -axis of the Euclidean distance between the membrane and the permeant (Fig. 1d), are shown for all enhanced-sampling methods in Fig. 5. Consistent with previous work²⁹, all PMFs exhibit a dominant free-energy barrier located at the bilayer center ($z = 0$ Å), where insertion of the hydrophilic DDA into the hydrophobic core incurs a cost of approximately 7–8 kcal/mol, along with two

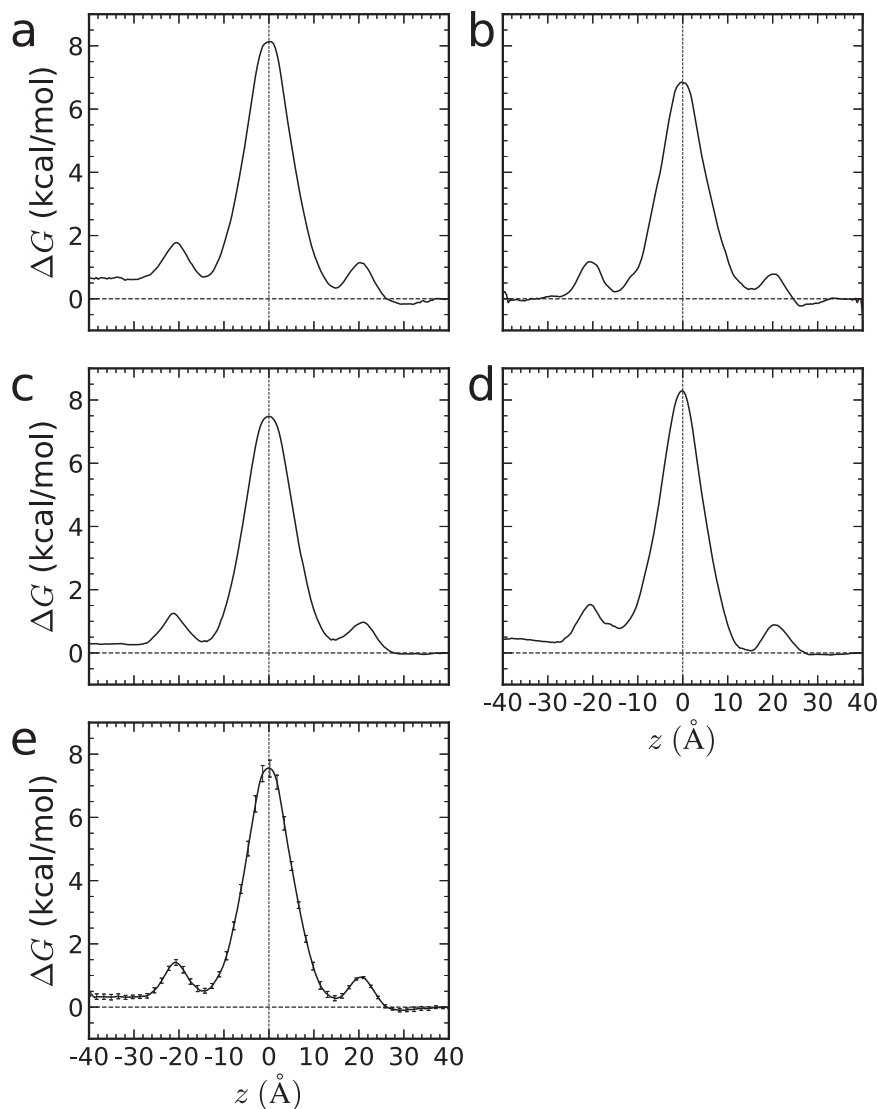


Fig. 5 | Membrane permeation: 2', 3'-dideoxyadenosine permeating a POPC bilayer (Fig. 1d). PMF determined along the normal, z , to the lipid bilayer using REUS (a), WT-MtD (b), WTM-eABF (c), OPES (d) and (e) average potential of mean

force computed from the marginal probabilities with error bars representing one standard deviation from the mean. Source data are provided as a Source Data file.

secondary barriers near the phospholipid head-group regions at $z = \pm 20$ Å, which reflect resistance from the interfacial environment. The REUS-derived PMF (Fig. 5a), obtained from ca. 5.0 μ s of sampling, reaches self-convergence after 0.3 μ s (Figure S7), with agreement relative to the marginalized average (Fig. 5e) achieved on the same timescale (Figure S7). It resolves a central barrier of approximately 8.0 kcal/mol, shoulders of ca. 1.0 kcal/mol near $z = \pm 20$ Å, and a persistent asymmetry of 0.8 kcal/mol between the two leaflets that does not dissipate with additional sampling (Figures S7 and S8). The WT-MtD PMF (Fig. 5b), generated from a 3.0- μ s simulation, attains self-convergence only after 2.8 μ s (Figure S7), with the RMSD relative to the marginalized average fluctuating between 0.5 and 1.0 kcal/mol before falling below $k_B T$ at the same time point (Figure S8). WT-MtD reproduces the interfacial barriers near $z = \pm 20$ Å at ca. 1.0 kcal/mol, but yields a reduced central barrier of 6.5 kcal/mol, and exhibits minimal hysteresis relative to the other methods. The WTM-eABF PMF (Fig. 5c), obtained after 6.2 μ s of sampling, reaches self-convergence after 1.7 μ s (Figure S7), while agreement with the marginalized average requires substantially longer sampling, approaching 4.0 μ s (Figure S8). The resulting profile recovers a central barrier of 7.5 kcal/mol and symmetric interfacial barriers of ca. 1.0 kcal/mol, with a reduced

shoulder asymmetry of 0.3 kcal/mol. Finally, the OPES-derived PMF (Fig. 5d), generated from a 3.0 μ s simulation using a barrier setting of 20 kcal/mol, reaches both self-convergence and agreement with the marginalized average on the same timescale (1.4 μ s; Figures S7 and S8), and reproduces a central barrier of 8.0 kcal/mol along with secondary barriers of 1.0 kcal/mol near $z = \pm 20$ Å, in quantitative agreement with the other methods. Additional OPES barrier settings of 10 and 15 kcal/mol yield qualitatively different PMFs characterized by larger hysteresis about $z = 0$ Å, despite satisfying the self-convergence criterion (Figures S7 and S8).

The marginalized average PMF for DDA permeation across a POPC bilayer (Fig. 5e) shows a central barrier of ca. 7.5-kcal/mol and interfacial barriers of ca. 1.0 kcal/mol near the headgroups with minimal error overall. The long MW WTM-eABF reference (Figure S11d) reproduces both the positions and magnitudes of these features, with an RMSD between the two PMFs of 0.3 kcal/mol. We note that the persistent asymmetry in the REUS PMF is consistent with incomplete relaxation of local membrane deformation within individual umbrella windows: because the time-independent biasing potential confines the permeant near a fixed z -position, the lipid packing perturbation induced near the headgroup regions may not fully equilibrate within

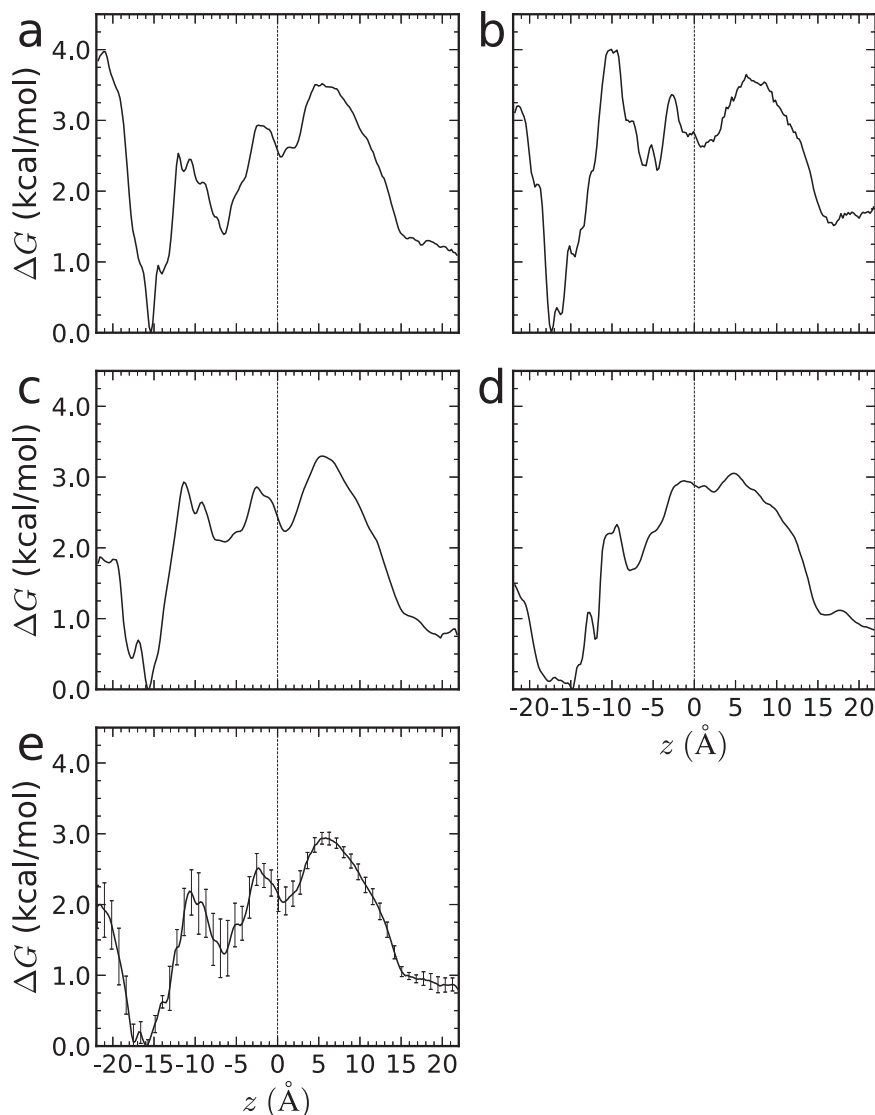


Fig. 6 | Ion permeation: Sodium permeating nAChR. Potential of mean force determined along the pore axis (Fig. 1e), using REUS (a), WT-MtD (b), WTM-eABF (c), OPES (d) and (e) average potential of mean force computed from the marginal

probabilities with error bars representing one standard deviation from the mean. Source data are provided as a Source Data file.

the sampling time of each window. In adaptive-biasing methods, the permeant traverses the full pathway repeatedly and samples different lateral positions in the membrane plane at each crossing, effectively averaging out local deformation artifacts and yielding more symmetric profiles.

nAChR channel permeation by sodium

The PMFs underlying sodium permeation through the nicotinic acetylcholine receptor (nAChR), characterized along z , coinciding with the long axis of the pore (Fig. 1e), are shown for all enhanced-sampling methods in Fig. 6. All PMFs are asymmetric with respect to the channel center ($z = 0$ Å) and exhibit a deep global minimum on the intracellular side near $z = -17$ Å, reflecting favorable electrostatic interactions between the sodium ion and the Glu260 residues lining the pore. This minimum is prefaced by a steep free-energy drop associated with the escape of the ion from the positively charged environment created by Arg50. As the ion progresses toward the extracellular side, three barriers are consistently observed: the first near $z = -10$ Å, the second near $z = -3$ Å, and a third, often dominant permeation barrier near $z = +6$ Å, where bulky Ile247 side chains restrict ion movement. These unfavorable interactions are partially mitigated by adjacent Ser248

residues. We note that the reaction-coordinate spans the pore interior only ($-22 \leq z \leq +22$ Å), with both endpoints lying within the channel rather than in bulk solvent on either side. Combined with the distinct residue environments encountered along the pore axis, this makes the PMF inherently asymmetric about the channel center, in contrast to the symmetric membrane permeation PMF of DDA. The REUS-derived PMF (Fig. 6a), obtained from a 0.4- μ s simulation, reaches self-convergence in less than 0.1 μ s (Figure S9), in agreement with the marginalized average (Fig. 6e) achieved on the same timescale (Figure S10). It reproduces the intracellular minimum at $z = -16$ Å and resolves a 2.5-kcal/mol barrier at $z = -10$ Å, a 3.0 kcal/mol barrier at $z = -5$ Å, and a gradual rise to a dominant 3.5 kcal/mol barrier near $z = 7.0$ Å, followed by a descent toward the extracellular vestibule. The WT-MtD PMF (Fig. 6b), generated from a 2.0- μ s simulation reaches self-convergence after 1.7 μ s (Figure S9), with the RMSD relative to the marginalized average stabilizing on the same timescale (Figure S10). WT-MtD resolves the same asymmetric profile and sequence of barriers, although the first barrier near $z = -10$ Å is exaggeratedly high at 4.0 kcal/mol, while the latter barriers at $z = -3$ and $+6$ Å remain consistent with other methods at 3.5 and 3.8 kcal/mol, respectively. The WTM-eABF PMF (Fig. 6c), obtained from a 1.5- μ s simulation, reaches

self-convergence after 0.7 μs (Figure S9), in agreement with the marginalized average achieved on the same timescale (Figure S10). It reproduces the intracellular minimum at $z = -17\text{ \AA}$, followed by successive barriers of 3.0 kcal/mol at $z = -10\text{ \AA}$ and $z = -3\text{ \AA}$, and a dominant barrier near $z = +7\text{ \AA}$ associated with steric and hydrophobic constriction by Ile247, after which the free energy decreases due to favorable electrostatic interactions with Glu241 residues at the extracellular vestibule. Last, the OPES-derived PMF (Fig. 6d), generated from a 2.0- μs simulation using a barrier setting of 5.0 kcal/mol, reaches self-convergence after 1.4 μs (Figure S9), while agreement with the marginalized average is achieved substantially earlier at 0.5 μs (Figure S10). OPES resolves the intracellular minimum near $z = -15\text{ \AA}$ and the three successive barriers at $z = -10, -3$, and $+5\text{ \AA}$, with the latter two barriers having heights of 2.9 and 3.1 kcal/mol, respectively, although the minimum in between is shallower than with the other methods. Increasing the OPES barrier height to 10 kcal/mol leads to slower convergence (1.9 μs ; Figure S9), but recovers the same PMF (Figure S10), indicating that OPES is robust for ion permeation provided that the barrier is not severely underestimated, while excessive overestimation may disrupt sampling due to rapid initial biasing.

For sodium permeation through the nAChR channel, the marginalized average PMF (Fig. 6e) reveals three barriers as the ion traverses the channel toward the extracellular side, with minimal errors overall. The long MW WTM-eABF reference (Figure S11e) is in close agreement, with an RMSD of ca. 0.4 kcal/mol relative to the marginalized average. We note that the pore-lining carboxylates near the deepest minimum ($z = -15\text{ \AA}$) can transiently bind spectator ions from solution, potentially introducing non-equilibrium effects. Thus, the PMFs reported represent a minimal estimate of the permeation free energy, and the presence of ions within the channel would be expected to yield larger apparent barriers.

Discussion

Landscape dominates method choice

Across the five systems examined, one organizing principle emerges with striking clarity. The dimensionality and topological complexity of the FEL determine method performance far more decisively than the choice of algorithm alone. Each of the four approaches examined here exhibits characteristic failure modes that are invisible in low-dimensional model systems, e.g., the alanine-dipeptide tests on which most methods are validated, but become consequential, and in some cases severe, in realistic biomolecular landscapes. This has a direct practical consequence: the most frequently cited advantages of adaptive-biasing methods over US-based approaches do not hold uniformly across problem types. For well-defined one-dimensional processes, the simpler, more transparent strategy wins, but for coupled, multidimensional transitions, the hybrid approaches earn their additional complexity. No method dominates all regimes, but the performance boundaries are sharp enough to support principled, prescriptive method selection, as outlined below.

REUS is the fastest and most reliable method for one-dimensional processes

For the three one-dimensional processes (dissociation of p41 from the Abl-SH3 domain, membrane permeation by DDA, and sodium permeation through nAChR), REUS converged faster than any other method, often by a substantial margin. For Abl-SH3:p41, REUS agreed with the marginalized average at ca. 0.2 μs , with WTM-eABF converging ever so slightly faster ($\leq 0.1\text{ \mu s}$), but WT-MtD required 0.3 μs and OPES 0.3–0.8 μs depending on the barrier setting. For membrane permeation, the contrast is greater, where REUS converged to the marginalized consensus in 0.3 μs , and WT-MtD required 2.8 μs , nearly a tenfold difference, and WTM-eABF reached agreement with the consensus average only at approximately 4.0 μs , despite self-converging earlier at 1.7 μs . For sodium permeation through nAChR,

REUS converged in under 0.1 μs compared to WT-MtD at 1.7 μs and WTM-eABF at 0.7 μs . These are not marginal differences.

For one-dimensional processes with physically meaningful, well-defined RCs, REUS consistently delivers the most reliable and computationally economical results. The common perception that its setup overhead renders it less practical than adaptive-biasing methods is not supported by these data. The simulation time saved can substantially outweigh the cost of window preparation. For this class of problems, REUS should be the default choice.

WT-MtD performed least efficiently on one-dimensional processes, showing the slowest convergence for membrane permeation and the largest residual deviations from the marginalized average for the ion permeation case, where it overestimated the first barrier near $z = -10.0\text{ \AA}$ by approximately 1.0 kcal/mol. This is consistent with the known tendency of WT-MtD to overfill free-energy minima before the biasing potential stabilizes, an effect amplified in systems with sharp, asymmetric features. For straightforward one-dimensional problems, the additional parameter tuning that WT-MtD demands does not translate into faster or more accurate results.

WTM-eABF is the most robust method for multidimensional landscapes

For the two multidimensional systems, ribose binding coupled to conformational rearrangement in RBP and the DFG-in/DFG-out transition in BCR-AblI kinase, the performance landscape shifts substantially. One result stands out: WT-MtD did not converge for the BCR-AblI kinase transition within the allocated simulation time. This is not a marginal failure. The DFG flip involves a rugged two-dimensional landscape with multiple low-lying minima separated by barriers of comparable magnitude, and the adaptive bias of WT-MtD proved insufficient to stabilize the FEL within 2.0 μs . Its RMSD relative to the final estimate remained above 2.0 kcal/mol for most of the trajectory, and the resulting free-energy difference between the DFG-in and DFG-out states (1.2 kcal/mol) deviates markedly from the near-isoenergetic values as reported in the literature²⁸. For RBP, WT-MtD did converge and yielded the lowest RMSD relative to the consensus average (approximately 1.5 kcal/mol), but required 2.2 μs , longer than either WTM-eABF or OPES under their optimal settings.

WTM-eABF was the most consistently reliable method for both two-dimensional systems. For the kinase, it reached self-convergence in 1.0 μs and agreement with the marginalized average (1.0–1.5 kcal/mol RMSD) within 0.4 μs , recovering a free-energy difference of 0.4 kcal/mol in favor of the DFG-in state, fully consistent with prior literature²⁸. For RBP, it resolved all three relevant conformational basins (HC, HO, AO) and reached self-convergence in 1.7 μs , though with elevated RMSD in high-energy regions. Crucially, WTM-eABF achieved this without requiring prior knowledge of barrier heights and without the recurrent instabilities observed with WT-MtD. The combination of statistically grounded force estimation from the eABF component with the exploratory barrier-filling of WT-MtD appears particularly well suited to landscapes with slow orthogonal degrees of freedom, precisely the conditions that define the most challenging cases examined here.

The performance of REUS on the two-dimensional systems warrants specific comment. For RBP, REUS self-converged after 1.5 μs but identified an AC state ($\theta = 114^\circ$, $r = 40\text{ \AA}$) rather than the AO state recovered by all other methods, maintaining a systematic RMSD of nearly 3.5 kcal/mol relative to the consensus average throughout the simulation. This reflects a systematic shift of the low-free-energy corridor attributable to bias introduced by the steered-MD structures used to seed the initial windows. The time-independent biasing potential within each window confines sampling to the vicinity of the window center, with inter-window communication restricted to replica-exchange moves, so that a poorly seeded corridor cannot self-correct. For the kinase transition, REUS converged correctly in terms of basin topology but yielded a free-energy difference of ca.

3.0 kcal/mol favoring the DFG-out state, again reflecting initial-seeding bias. For multidimensional problems, REUS, therefore, requires substantially more careful initialization, and its sensitivity to the quality of initial structures, compounded by the requirement of tightly synchronized parallel execution, makes it vulnerable to hardware availability, job scheduling, and replica failures in ways that adaptive-biasing methods are not.

OPES: rapid convergence and the risk of barrier mis-specification

OPES occupied a distinct position in this evaluation, not because it underperformed consistently, but because its failure mode is qualitatively different from those of the other methods and carries greater practical risk. When the barrier setting was well chosen, OPES performed competitively or better. For membrane permeation at a barrier of 20.0 kcal/mol, it converged in 1.4 μ s, faster than both WT-MtD (2.8 μ s) and WTM-eABF (1.7 μ s). For the kinase transition at 10.0 kcal/mol, it reached agreement with the consensus average at 0.8 μ s and for ion permeation at 5 kcal/mol, it did so at 0.5 μ s.

The critical issue is what happens when the barrier setting is incorrect. For the Abl-SH3:p41 system, barrier settings of 18.0 and 25.0 kcal/mol both satisfied the self-convergence criterion, where the RMSD relative to the final PMF plateaued below $k_B T$, yet, the resulting PMFs were inconsistent with all other methods and with the consensus average. For membrane permeation, barrier settings of 10.0 and 15.0 kcal/mol produced PMFs with qualitatively incorrect hysteresis about the bilayer center, again while appearing self-consistent by every standard diagnostic. This is the defining risk of OPES. A simulation can appear converged by every standard internal diagnostic while producing a quantitatively incorrect free-energy profile. For WT-MtD, sub-optimal parameters lead to slow convergence that is manifest in the RMSD time series. For REUS, initialization failures produce deviations perhaps diagnosable through histogram inspection. For OPES, the standard self-convergence metric can issue a false certificate of quality.

This does not disqualify OPES, as its speed advantage under optimal conditions is real, but it imposes a non-negotiable validation requirement. Systematic testing of multiple barrier settings must be treated as part of the method itself, not as an optional check. Self-convergence alone is not a sufficient stopping criterion. Cross-method agreement or comparison to an independent, extended-timescale reference simulation should always accompany OPES-derived free-energy profiles, particularly for new systems where the relevant barriers are not known in advance.

A decision framework for method selection

The preceding sections reveal that each method occupies a distinct performance regime defined more by problem characteristics than by algorithmic sophistication. Table 2 distills this evidence into actionable recommendations.

Three points merit emphasis beyond the table. First, for exploratory studies where the landscape topology is unknown, a two-stage workflow is recommended: an initial WT-MtD or WTM-eABF

simulation to map coarse topography, followed by targeted WTM-eABF or REUS calculations to refine quantitative free-energy differences. This leverages WT-MtD's genuine strength of rapid barrier crossing and broad exploration while avoiding its limitations as a standalone quantitative method. A natural extension of this idea is to combine two or more enhanced-sampling algorithms to mitigate the individual limitations identified here. Indeed, WTM-eABF is itself a hybrid, and its effectiveness on the multidimensional systems considered here illustrates both the potential of such combinations and their cost, as the method introduces additional parameters governing the balance between the eABF and WT-MtD components and the extended-system coupling. More elaborate combinations have been proposed. For instance, coupling REUS with solute tempering (REST2³⁰ or gREST³¹) can alleviate the need to define low-dimensional collective variables by enhancing conformational exchange along orthogonal degrees of freedom; bias-exchange metadynamics³² distributes different CVs across replicas, extending WT-MtD to higher-dimensional landscapes; and OneOPES³³ layers OPES variants of increasing exploratory intensity within a replica-exchange framework, potentially reducing the sensitivity to barrier mis-specification identified here. In each case, however, the gains come at the cost of a larger replica count, a broader parameter space requiring joint optimization (as in the case of WTM-eABF identified here), and tighter requirements on inter-replica communication. All of which becomes increasingly demanding for standalone GPU nodes that are widely used and favor methods not requiring synchronized parallel execution.

Second, these results underscore a broader methodological principle, that self-consistency, i.e., the internal stability of a single free-energy estimate over time, is a necessary but not sufficient condition for correctness. In the limit of infinite sampling, all four methods must recover the same FEL, though, on any finite timescale, there is no guarantee that they will. Cross-method agreement and comparison to independent, extended-timescale reference simulations therefore provide the only reliable evidence that a reconstructed FEL reflects the true underlying thermodynamics. The results presented here show that self-convergence and cross-method agreement can diverge substantially, and that this divergence is most acute precisely for the methods and parameter regimes most likely to be chosen by practitioners seeking rapid convergence.

Third, the increasing integration of enhanced-sampling algorithms into major MD engines including AMBER³⁴, CHARMM³⁵, GENESIS³⁶, GROMACS³⁷, NAMD³⁸, and OpenMM³⁹, together with workflow tools such as the CHARMM GUI Enhanced Sampler⁴⁰ and the binding free-energy estimator BFEE2⁴¹, has substantially lowered the barrier to entry for these calculations. While this democratization is essential for the broader adoption of these methods in drug discovery and molecular design, it also amplifies the need for rigorous validation protocols, since many users may rely on default convergence diagnostics that, as we show here, can be insufficient. The benchmark framework presented here is intended to complement these accessibility efforts by providing a concrete basis for evaluating method reliability across biologically relevant complexity. Taken together,

Table 2 | Prescriptive guidance for enhanced-sampling method selection based on the present work

Problem class	Recommended method	Key caveat
One-dimensional, well-defined RC model (binding, permeation, ion conduction)	REUS	Substantial setup cost; sensitive to initial seeding
Two-dimensional or coupled conformational transition	WTM-eABF	Requires careful tuning of eABF/WT-MtD balance and extended-system coupling
Exploratory study / unknown FEL topology	WT-MtD or WTM-eABF	WT-MtD fragile on rugged landscapes; use as first-pass mapping only, followed by more quantitative methods
Rapid convergence needed, barrier estimate available	OPES	Always test multiple barrier settings; self-convergence alone is insufficient; cross-method validation mandatory

these three considerations argue directly for a change in community practice.

These considerations translate into a concrete workflow (Table 2): REUS as the default for one-dimensional processes, where it converged fastest, often by nearly an order of magnitude; WTM-eABF for coupled or multidimensional transitions, including the rugged BCR-Abl1 kinase DFG flip on which WT-MtD failed to converge altogether; OPES where an approximate barrier estimate is available, with systematic multi-barrier testing treated as integral to the protocol; and a two-stage exploration-then-refinement strategy for landscapes of unknown topology. Under no circumstances should self-convergence alone serve as the sole stopping criterion. Cross-method validation should become a standard component of enhanced-sampling free-energy calculations, reported alongside self-convergence metrics as a matter of methodological rigor. Doing so would transform method failures from silent and insidious into visible and correctable. By delineating performance regimes, limitations, and characteristic failure modes across realistic biomolecular systems, this study lays a practical foundation for more informed method selection, more reproducible free-energy calculations, and the future development of hybrid workflows that deploy complementary methods within a unified, cross-validating framework.

Methods

Enhanced-sampling algorithms

Replica-exchange umbrella sampling. Replica-exchange umbrella sampling (REUS) is a type of generalized ensemble method in which multiple replicas of a system, each restrained by a potential along one or more chosen CVs, are simulated in parallel. These replicas (also referred to as windows) periodically exchange configurations according to a Metropolis criterion to enhance sampling and improve overlap between individual windows^{9,17}. The restraining potentials V_{bias}^i typically take the form

$$V_{\text{bias}}^i(\mathbf{x}) = \frac{k_i}{2} (\xi(\mathbf{x}) - \xi_i)^2 \quad (2)$$

where $\xi(\mathbf{x})$ is the chosen CV, ξ_i is the center of the restraint in the i th window, and k_i is the force constant. The Metropolis criterion gives the probability p_{ij} of exchange and takes the form

$$\begin{cases} p_{ij} &= \min(1, \exp(-\beta(\Delta_1 + \Delta_2))) \\ \Delta_1 &= V_{\text{bias}}^i(\mathbf{x}_j) - V_{\text{bias}}^i(\mathbf{x}_i) \\ \Delta_2 &= V_{\text{bias}}^j(\mathbf{x}_i) - V_{\text{bias}}^j(\mathbf{x}_j) \end{cases} \quad (3)$$

where $\Delta_{1,2}$ give the change in energy upon swapping the biases i and j , and $\beta = 1/k_B T$. If the energy upon swapping is negative (favorable), the replicas will always exchange; otherwise, they exchange with probability given by the Boltzmann factor.

The output of an REUS run is a set of trajectories and CV values, which first have to be sorted into the original windows due to the swaps between them during the run. Histograms on the CV(s) for each window reveal the degree to which they overlap with one another, a necessary condition for convergence. Approaches to derive the free energy from the histograms include MBAR⁴² and WHAM⁴³. WHAM iteratively solves for the maximum-likelihood estimates of the free energies that make all window histograms self-consistent, weighting each data point by how reliable it is given its bias and sampling statistics. Once converged, WHAM yields the probability distribution $p(\xi)$ along the CV, from which the free energy is obtained via the equation $F(\xi) = -1/\beta \ln p(\xi)$.

Well-tempered metadynamics. The well-tempered metadynamics (WT-MtD) method belongs to the family of adaptive-biasing

techniques, where a history-dependent bias potential is added to the system to accelerate sampling along the chosen CV, $\xi(\mathbf{x})$. The bias potential at simulation step n is expressed as

$$V_n(\xi) = U^0(\xi) + \sum_{k < n} w_k G(\xi, \xi_k), \quad (4)$$

where $U^0(\xi)$ is an arbitrary reference function, w_k is the height (or weight) of the k -th Gaussian hill, and $G(\xi, \xi_k)$ is a multidimensional Gaussian kernel centered at the previously visited CV value $\xi_k = \xi(t_k)$, defined as

$$G(\xi, \xi_k) = h \exp \left[-\frac{(\xi - \xi_k)^2}{2\sigma^2} \right], \quad (5)$$

Here, $h = (\sigma\sqrt{2\pi})^{-1}$ is the normalization constant, and σ is the kernel width controlling the spread of the Gaussian along the CV direction. The prefactor w_k represents the Gaussian height, which in the well-tempered formulation is adaptively scaled according to

$$w_k = h_0 \exp[-\beta' V_k(\xi_k)], \quad (6)$$

where h_0 is the initial Gaussian height and β'^{-1} is an effective bias temperature. In the limit $\beta' \rightarrow 0$, WT-MtD reduces to conventional metadynamics with a constant deposition rate.

At long times, the ensemble-averaged bias converges to a rescaled estimate of the underlying FEL, $A(\xi)$,

$$\langle V_n(\xi) \rangle_{t \rightarrow \infty} \simeq -\frac{1}{1 + \beta'/\beta} A(\xi) + u(t), \quad (7)$$

where $u(t)$ is an additive constant, and $A(\xi)$ represents the PMF along the CVs, which can be recovered, up to an additive constant, by reweighting the biased ensemble. The term $U^0(\xi)$ can be taken as a flat baseline or an initial guess for the FEL to accelerate convergence, although the final estimate of $A(\xi)$ is independent of this choice. It is worth noting that in the current implementation in Colvars^{44,45}, the PMF is estimated by summing up the Gaussian hills and scaling the results on all grid bins with a scaling factor determined by the bias temperature. The reader is referred to alternate reweighting schemes to improve convergence of the biasing potential^{46,47}.

Well-tempered metadynamics-extended adaptive-biasing force.

The well-tempered metadynamics-extended adaptive-biasing force (WTM-eABF) method is based on the principle of combining two distinct enhanced-sampling techniques to accelerate the convergence of free-energy calculations^{19,48}. This hybrid approach leverages the strengths of both eABF^{49,50}, which effectively shaves free-energy barriers, and well-tempered MtD¹⁸, which fills free-energy valleys. The synergy between these two methods leads to an efficient exploration of the conformational space. In the WTM-eABF framework, fictitious particle, ξ' , is introduced into the system. This particle is harmonically coupled to the CV of interest, $\xi(\mathbf{x})$. The extended potential energy of the system is defined as:

$$\begin{cases} U'(\xi', \mathbf{x}) &= U(\mathbf{x}) + V'[\xi', \xi(\mathbf{x})] \\ V'[\xi', \xi(\mathbf{x})] &= \frac{1}{2} k [\xi' - \xi(\mathbf{x})]^2 \end{cases} \quad (8)$$

where $U(\mathbf{x})$ is the original potential energy of the system and $V'[\xi', \xi(\mathbf{x})]$ is the harmonic coupling potential. The term k represents the force constant of the spring that connects the fictitious particle, also known as the extended CV, to the original CV. A key feature of this method is that both the ABF and MtD biases are applied directly to the fictitious particle, ξ' , rather than to the original CV. The total biasing force in WTM-

eABF is therefore a sum of the ABF and the WT-MtD components:

$$\begin{cases} F_{\text{WT-MtD-eABF}} = F_{\text{ABF}} + F_{\text{WT-MtD}} \\ F_{\text{ABF}} = k(\xi' - \xi(\mathbf{x}))_{\xi'} \\ F_{\text{WT-MtD}} = -\frac{\partial}{\partial \xi'} \left[\sum_{m < n} w_m \exp \left(-\frac{(\xi' - \xi_m)^2}{2\sigma^2} \right) \right] \end{cases} \quad (9)$$

In these equations, F_{ABF} is the adaptive-biasing force derived from the harmonic spring, and $F_{\text{WT-MtD}}$ is the force from the WT-MtD potential. The MtD potential is constructed from a history-dependent sum of Gaussians, where n is the number of Gaussians deposited, w_m is the history-dependent height, and σ determines the width. Once the simulation converges, the unbiased free-energy profile along ξ can be reconstructed. Several estimators are available for this purpose, including the corrected z-averaged restraint (CZAR)⁵⁰, umbrella integration (UI)^{51,52}, and the MBAR method^{42,53,54}. For example, the CZAR estimator is expressed as:

$$\frac{dA(\xi)}{d\xi} = -\frac{1}{\beta} \frac{d[\ln \tilde{p}(\xi)]}{d\xi} + k(\langle \xi' - \xi \rangle_{\xi}) \quad (10)$$

where $\tilde{p}(\xi)$ is the biased marginal probability distribution of the original CV, ξ . The first term on the right-hand side is the standard free-energy gradient from the biased distribution, while the second term serves as a correction that removes the average work done by the harmonic spring, thereby accounting for the bias introduced by the coupling potential.

On-the-fly probability enhanced sampling. It has been shown that the FEL obtained via reweighting^{46,47,55} can converge faster than the biasing potential itself⁵⁶. Rather than directly constructing the biasing potential by accumulating the Gaussian hills as in MtD and its variants, the OPES^{21,57} estimates the probability distribution of CVs via reweighting and then defines the biasing potential from it. In OPES, the biasing potential at step n is computed as

$$V_n(\xi) = -\frac{1}{\beta} (1 - 1/\gamma) \ln \left(\frac{p_n(\xi)}{Z_n} + \epsilon \right) \quad (11)$$

where γ is the bias factor, and ϵ is the regularization constant for the probability. γ and ϵ are determined by the estimated height of the free-energy barrier to overcome. The probability $p_n(\xi)$ and the normalization factor Z_n are defined as

$$\begin{cases} p_n(\xi) &= \frac{\sum_{k=0}^n w_k G(\xi, \xi_k)}{\sum_{k=0}^n w_k} \\ Z_n &= \frac{1}{|\Omega_n|} \int_{\Omega_n} p_n(\xi) d\xi \\ w_k &= \exp(\beta V_{k-1}(\xi_k)) \\ G(\xi, \xi_k) &= \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{(\xi - \xi_k)^2}{2\sigma^2} \right] \end{cases} \quad (12)$$

where $|\Omega_n|$ is the volume of the explored CV space at step n , σ is the parameter controlling the width of the Gaussian hill $G(\xi, \xi_k)$. OPES requires only a few parameters to start, namely the estimated height of the free-energy barrier to overcome (to determine γ and ϵ), the frequency of updating the biasing potential, and the initial bandwidth of the Gaussian hill.

OPES enables a simple and straightforward reweighting scheme to recover the unbiased probability distribution and the associated PMF along ξ . Assuming that the PMF is discretized onto a histogram, and ξ' is the value of CV at any one of the grid centers of the histogram, the

PMF at ξ' , $A(\xi')$ can be computed as

$$\begin{cases} \langle p(\xi') \rangle &= \frac{\sum_{t=0}^N (\xi(\mathbf{x}) - \xi') \exp(\beta V_t(\xi'))}{\sum_{t=0}^N \exp(\beta V_t(\xi'))} \\ A(\xi') &= -k_B T \ln \langle p(\xi') \rangle + A_0 \end{cases} \quad (13)$$

where A_0 is a constant to ensure that $A(\xi')$ is non-negative for any ξ' . OPES has been ported from PLUMED⁵⁸ to Colvars (version 2024-10-16)^{44,45}, which has been integrated into NAMD 3.1alpha2³⁸.

Computational assays

Protein-ligand binding Abl-SH3:p41. The standard binding free energy can be estimated using a stepwise geometrical approach^{59,60}. Within this procedure, the simulation of the physical separation of the ligand from the protein represents the most critical and computationally intensive stage. This process is guided by geometrical restraints applied to the spherical-coordinate angles, Euler angles, and optionally, the distance RMSD of the ligand backbone atoms^{24,59}. Given its computational demands, this dissociation step serves as an ideal computational assay to test the efficiency and reliability of each enhanced-sampling algorithm. We use the Abl-SH3:p41 complex (PDB ID: 1BBZ) for this test because it is a well-characterized benchmark with established force-field accuracy^{24,59-61}. The initial structure of the system is modeled using CHARMM-GUI⁶². The all-atom CHARMM22/CMAP force field^{63,64} is employed for both the protein and the decapeptide ligand, while the water molecules are described by the TIP3P model⁶⁵. The final simulation system contained 36,311 atoms, including 11,757 water molecules, within a cubic box of ca. 71 Å. Prior to the production runs, the complex structure was pre-equilibrated for 0.1 μs. All input files for the simulations are generated using the binding free-energy estimator (BFEE) tool^{24,41,66}, with minor manual modifications made to select the desired enhanced-sampling algorithm. All simulations were performed in the isothermal-isobaric ensemble.

Coupled protein-ligand binding and conformational transition of the ribose-binding protein. For the RBP holo simulations, the X-ray structure of holo RBP in complex with ribose (PDB ID: 1DRJ) was used as the initial model²⁵. The setup followed reference²⁶. All histidine residues were kept neutral, except for His152. The protein was solvated in a cubic water box using CHARMM-GUI⁶⁷. The final simulation system contained 53,641 atoms, including 16,515 water molecules, within a cubic box of approximately 81 Å per side. The system was first energy-minimized for 10,000 steps using the conjugate gradient method prior to equilibration. Subsequently, the system was gradually relaxed using constrained MD simulations for 1 ns following the standard CHARMM-GUI protocol⁶⁷. All simulations were performed using the CHARMM36m force field^{68,69}. The initial equilibration was conducted for 10 ns and the production simulations were carried out in the isothermal-isobaric ensemble.

Conformational transition during BCR-Abl1 kinase DFG-in/DFG-out activation. The initial structure of the apo BCR-Abl1 kinase was provided by the authors of reference²⁸. Briefly, in that work, the starting structure was PDB ID: 5MO4 with missing parts added from PDB ID: 1OPL. Additionally, the T315I mutation in the structure was reverted to Thr, but the D363N mutation was retained²⁸. The solvated system contained 28,855 water molecules within a cubic box of ca. 100 Å, and was neutralized and ionized to 150 mM sodium chloride. The CHARMM36m force field was used for the protein⁶⁸, with the TIP3P model⁶⁵ for water. The system underwent 60 ns of equilibration prior to free-energy calculations.

Membrane permeation by 2', 3'-dideoxyadenosine. 2', 3'-dideoxyadenosine is a dideoxynucleoside in which the 3'-hydroxyl group of the ribose moiety is replaced by a hydrogen atom. The force-field parameters for the permeant were generated using the CHARMM general force field (CGenFF) program⁷⁰, yielding negligible penalties. To validate these parameters, free-energy perturbation (FEP) calculations were carried out to evaluate the relative solvation free energy in water and n-decane. The permeant was then inserted into a pure 1-palmitoyl-2-oleoyl-glycero-3-phosphocholine (POPC) lipid bilayer, which had been trimmed from a larger pre-equilibrated membrane and further equilibrated for 100 ns. The final system comprised 64 lipids per leaflet, sufficient to accommodate local undulations induced by the permeant. The membrane normal was aligned along the *z*-axis, with each leaflet in contact with a water layer at least 32-Å thick. The equilibrated simulation box measured ca. $67 \times 67 \text{ Å}^2$ in the *x, y*-plane and 90 Å along the *z*-axis. Steered molecular dynamics (SMD) simulations⁷¹ were subsequently performed to pull the permeant across the membrane in a near-equilibrium regime at a constant velocity of 5 Å/ns, for a total of 0.1 μs. To improve sampling efficiency, the reaction pathway was divided into nine windows, corresponding to different initial *z* positions. Each window was independently equilibrated while restraining the permeant near its central position by means of a harmonic potential.

nAChR channel permeation by sodium. The computational assay was built, starting with the crystal structure of the human $\alpha\beta_4$ nicotinic acetylcholine receptor in complex with nicotine (PDB ID: 6PV7), considered here in its apo form. Using CHARMM-GUI⁷², the protein was inserted in a fully hydrated membrane composed of 300 POPC and 56,836 water molecules. Sodium chloride was added to achieve a physiological concentration of 150 mM, while ensuring electric neutrality of the assay. The resulting molecular assembly consisted of 242,838 atoms placed in a rhombohedral cell of approximately $110 \times 110 \times 142 \text{ Å}^3$ after adequate equilibration. The all-atom CHARMM36 force field was utilized to describe the protein, the lipids, and the ions. For water, use was made of the TIP3P model⁶⁵. For a subset of heavy atoms of both the protein and the lipids, part of the mass was transferred onto the covalently bonded hydrogen atoms, resulting in an effective hydrogen mass of 3.024 g/mol^{73,74}. As a preamble to the enhanced-sampling simulations describing the

permeation of the nAChR channel by sodium, the system was first thermalized with the positions of the heavy atoms of the protein restrained harmonically, prior to a long 100-ns equilibration bereft of geometric restraints.

Free-energy calculations

All the simulations described hereafter were performed using the NAMD 3.0 program³⁸ with the Colvars library^{44,45}. Water molecules were constrained using the SETTLE algorithm⁷⁵, while all remaining bonds involving hydrogen atoms were constrained using the RATTLE algorithm⁷⁶. The equations of motion were integrated with the Verlet r-RESPA multiple time-step scheme⁷⁷, and long-range electrostatic interactions were treated using the particle-mesh Ewald (PME) method⁷⁸. Specific simulation parameters including integration time step, non-bonded cut-offs and switching distances, temperature and pressure coupling schemes, PME grid spacing, and multiple-time-step settings are summarized in Table 3 for each process.

Protein-ligand binding Abl-SH3:p41. The CV utilized in the enhanced-sampling simulations of the Abl-SH3:p41 complex, denoted *r*, was defined as the distance between the center of mass of the p41 peptide and that of the Abl-SH3 domain. All remaining geometric degrees of freedom were restrained as described in the Computational Assays subsection. REUS simulations employed 36 windows spanning the range $10.3 \leq r \leq 34.0 \text{ Å}$. To ensure sufficient overlap between neighboring windows, the spacing was refined to 0.5 Å in the interaction region $12.0 \leq r \leq 23.0 \text{ Å}$, and set to 1.0 Å elsewhere. Each window was sampled for 17.3 ns, yielding an aggregated simulation time of approximately 0.62 μs. Replica-exchange attempts were performed every 2 ps, resulting in an average acceptance probability of 21%. WT-MtD simulations were performed using an initial Gaussian hill height of 0.05 kcal/mol, a Gaussian width of 3.0 Å ($2\sigma = 3 \text{ Å}$), a bias temperature of 4000 K, and a hill deposition interval of 2.0 ps. A single replica was simulated for a total of 0.3 μs. The PMFs were reconstructed using the reweighting procedure implemented in the Colvars library^{44,45}. WTM-eABF simulations were performed using a threshold of 10,000 samples to apply the biasing force. The metadynamics component employed a Gaussian width of 0.6 Å, a hill weight of 0.05 kcal/mol, and a bias temperature of 4000 K. The total simulation time was 300 ns. Free-energy reconstruction was carried out using the CZAR estimator⁵⁰. The

Table 3 | Simulation parameters for each biomolecular system

Parameter	Biomolecular system				
	Abl-SH3:p41	RBP	BCR-Abl1	DDA	nAChR
Time step (fs)	2.0	2.0	2.0	2.0	4.0 ^a
Nonbonded ^b update (steps)	1	1	1	1	1
Electrostatics ^c update (steps)	2	2	2	2	4
Thermostat	Langevin	Langevin	Stoch. rescal. ^d	Stoch. rescal. ^d	Stoch. rescal. ^d
Thermostat temperature (K)	300.0	310.0	300.0	308.0	310.0
Langevin damping (ps ⁻¹)	1.0	1.0	–	–	–
Stochastic rescale period (steps)	–	–	1	1	1
Barostat	Lang. piston ^e	Lang. piston ^e	Lang. piston ^e	Lang. piston ^e	Lang. piston ^e
Target pressure (bar)	1.01325	1.01325	1.01325	1.01325	1.01325
Piston period (fs)	200.0	200.0	300.0	400.0	400.0
Piston decay (fs)	100.0	100.0	150.0	300.0	300.0
Nonbonded cutoff (Å)	12.0	12.0	12.0	12.0	12.0
Switching distance (Å)	10.0	10.0	10.0	10.0	10.0
PME grid spacing (Å)	1.0	1.0	1.2	1.2	1.2

^a Using hydrogen-mass repartitioning^{73,74}.

^b Short-range contribution of electrostatic and van der Waals interactions.

^c Long-range contribution of electrostatic interactions.

^d Using stochastic velocity rescaling⁸⁵.

^e Using the Langevin piston algorithm⁸⁶.

long MW WTM-eABF reference was run for 3.6 μ s. OPES simulations were performed with a Gaussian deposition stride of 2 ps. To assess sensitivity to the barrier parameter, four barrier heights were tested, namely 18, 20, 22.5, and 25 kcal/mol. For each barrier setting, two independent 1.2- μ s replicas were performed. Adaptive Gaussian width schemes were enabled to dynamically adjust kernel sizes according to local fluctuations of the CV. A neighbor list was employed to accelerate bias evaluation.

Coupled protein-ligand binding and conformational transition of the ribose-binding protein. Two CVs were employed to describe the coupling between ribose binding and conformational rearrangement in RBP. The first CV was the hinge angle, θ , defined as the bending angle formed by the centers of mass of the N-terminal domain (NTD; residues 1–100 and 236–259), the C-terminal domain (CTD; residues 108–231 and 269–271), and the hinge region (residues 101–107, 232–235, and 260–268). The second CV was the ribose-binding-site distance, r , defined as the distance along the z -axis between the center of mass of all ribose atoms and the $C\alpha$ atom of residue Ser103. To suppress global translation and rotation, the center of mass of all $C\alpha$ atoms of RBP, together with their Euler angles, was restrained to the values obtained from the final conformation of the equilibrium simulation using harmonic restraints with a force constant of 100 kcal mol^{−1} Å^{−2} (degree²). In addition, the distance between the center of mass of the ribose and the $C\alpha$ atom of Ser103 orthogonal to the z -axis was restrained to remain below 4 Å using a harmonic potential with a force constant of 500 kcal mol^{−1} Å^{−2}. Initial conformations for the REUS simulations were generated using a two-stage SMD protocol⁷¹. Starting from the final structure of the equilibrium simulation, a 10-ns SMD trajectory was first performed in which the hinge angle θ was pulled from 100° to 160° while restraining the ribose distance r at 10 Å. Harmonic force constants of 200 kcal mol^{−1} degree^{−2} and 100 kcal mol^{−1} Å^{−2} were applied to θ and r , respectively. From this trajectory, 22 configurations were extracted at hinge angles spanning 100° ≤ θ ≤ 160°, spaced by approximately 2.7°. From each of these configurations, additional SMD simulations were carried out in which the ribose distance r was pulled from 10 Å to 40 Å, while maintaining θ fixed at its initial value using the same force constants. This two-step procedure enabled coverage of the full (θ , r) configurational space. The resulting CV space was discretized onto a 22 × 22 grid (484 windows), corresponding to resolutions of approximately 1.4 Å in r and 2.7° in θ . Two-dimensional REUS simulations were then performed using these grid centers. Each replica was simulated for 3.8 ns, yielding an aggregate sampling time of approximately 1.8 μ s. Harmonic restraints with force constants of 0.5 kcal mol^{−1} degree^{−2} and 0.5 kcal mol^{−1} Å^{−2} were applied to θ and r , respectively. Replica-exchange attempts between the four nearest-neighbor windows were carried out every 1 ps, resulting in an average exchange acceptance of approximately 5%. The first 1 ns of each trajectory was discarded as equilibration. The free-energy surface was reconstructed using a non-parametric reweighting procedure^{42,79}, conceptually equivalent to MBAR⁸⁰. WT-MtD simulations were performed using a hill deposition stride of 2.0 ps, with initial Gaussian heights of 0.1 kcal/mol and widths of 0.2° and 0.1 Å for θ and r , respectively. The bias temperature was set to 1500 K. A single replica was simulated for 3.0 μ s, and the resulting two-dimensional FELs were reconstructed using the reweighting procedure implemented in the Colvars library^{44,45}. WTM-eABF simulations employed an extended-Lagrangian formulation with fluctuation amplitudes of 0.4° for θ and 0.1 Å for r , and coupling time constants of 200 fs for both CVs. A minimum of 20,000 samples per bin was required prior to bias application. In parallel, well-tempered metadynamics was applied along both CVs using Gaussian hills with widths of 2.0° and 1.0 Å, deposited every 2.0 ps with an initial height of 0.3 kcal/mol. The bias factor corresponded to a bias temperature of 1500 K. Free energies were reconstructed using the CZAR estimator⁵⁰, and the

total simulation time was 2.5 μ s. The long MW WTM-eABF reference was run for 5.1 μ s. OPES simulations were performed using barrier settings of 10, 20, and 30 kcal/mol. Gaussian hills were deposited every 2.0 ps with widths of 2.5° and 1.0 Å for θ and r , respectively. A neighbor-list-accelerated OPES kernel was employed, and the evolving FEL was reconstructed on-the-fly. Each barrier setting was simulated for 2.9 μ s.

Conformational transition during BCR-Abl1 kinase DFG-in/DFG-out activation. Exploration of the two-dimensional FEL underlying the DFG in/out conformational transition in the apo BCR-Abl1 kinase was achieved using two dihedral angles centered on Phe382 of the DFG motif. Specifically, ψ , defined by atoms C_{β} – C_{α} –C(Phe382)–N(Gly383), and ϕ , defined by C_{β} – C_{α} –N(Phe382)–C(Asp381). The use of the C_{β} atom in both definitions provides a more robust angular reference and ensures smooth sampling of the transition between the DFG-in and DFG-out states. In REUS simulations, the (ψ , ϕ) CV space was discretized into a 36 × 36 grid spanning −180° to +180° for both variables (10° spacing), resulting in 1,296 umbrella windows. Initial configurations were generated by two-dimensional SMD simulations starting from the equilibrated DFG-out conformation: First, ϕ was restrained at 115° while ψ was pulled from 180° to −180°; subsequently, for each ψ value, ψ was fixed while ϕ was pulled from 180° to −180° using harmonic restraints ($k_{\text{pull}} = 0.1$ and $k_{\text{fix}} = 0.5$ kcal · mol^{−1} · degree^{−2}). The resulting structures were used as starting configurations for the REUS simulations. Each umbrella window was pre-equilibrated for 1 ns under stronger restraints ($k = 0.5$) and then sampled for 3.8 ns under weaker restraints ($k = 0.01$), totaling 4.9 μ s across all windows. Replica exchanges between neighboring windows were attempted every 4 ps with an average acceptance ratio of 18.8%. The final two-dimensional FEL was reconstructed using WHAM^{43,81}. WT-MtD employed Gaussian hills with widths of 2.0° for both dihedrals and an initial hill height of 0.1 kcal/mol, deposited every 2 ps, with a bias temperature of 4000 K. For WTM-eABF, the reaction coordinate was discretized into bins 5.0° wide over the interval −180 ≤ z ≤ 180 Å, with harmonic wall restraints applied at the boundaries using a force constant of 100 kcal mol^{−1} Å^{−2}. The extended variable was coupled to the physical coordinate with a fluctuation amplitude of 1.0° and a coupling time constant of 100 fs, and the mean force was accumulated only after a minimum of 5000 samples per bin had been collected. In parallel, well-tempered metadynamics used Gaussian hills of width 15° and an initial height of 0.1 kcal/mol, deposited every 2.0 ps with a bias temperature of 3000 K. Free energies were reconstructed using the CZAR estimator. The long MW WTM-eABF reference was run for 12.1 μ s. The OPES simulations utilized Gaussian kernels of 5° for both CVs, which were deposited every 2.0 ps, and simulations for the barrier values of 7 and 10 kcal/mol were run. For both barrier settings, the widths were dynamically set based on the variance of the CVs and a neighbor list was used for kernel summation. The PMF as a function of angles ψ and ϕ were computed on the fly by reweighting. The simulation time run for each barrier setting was 2.0 μ s.

Membrane permeation by 2', 3'-dideoxyadenosine. The RC model describing the permeation of DDA across the membrane event was defined as the projection along its normal, that is, the z -axis, of the Euclidean distance between the centers of mass of the permeant and of the lipid bilayer. This coordinate, thus, tracks the relative displacement of the permeant as it crosses the membrane, with $z = 0$, corresponding to the bilayer midplane, and positive and negative values indicating motion toward either aqueous phase. The entire permeation pathway extends over approximately 90 Å, i.e., −40 ≤ z ≤ +40 Å, thereby encompassing both bulk-water regions and the hydrophobic core of the bilayer. Depending upon the enhanced-sampling algorithm employed, and to test the efficiency of the latter, either no stratification was used, or the pathway was divided into a limited number of

overlapping strata, each spanning 30 Å, with a 5 Å overlap between consecutive windows. Within each stratum, the reaction-coordinate model was discretized into bins 0.1 Å wide. The REUS simulation consisted of 81 harmonic windows stretching over the permeation coordinate with even spacing. The force constant on each window was 2.5 kcal/mol Å². The initial window configurations were extracted from the WTM-eABF trajectory. To ensure the membrane stays centered in the periodic box, harmonic restraints were applied to the phosphate atoms with a force constant of 5.0 kcal mol⁻¹ Å⁻². Exchanges between neighboring replicas were attempted every 5 ps. The total simulation time over all umbrella windows was 5.2 μs. The WT-MtD simulation was performed with Gaussian hills of width 5.0 Å and initial height 0.1 kcal/mol, deposited at fixed intervals of 2.0 ps under a well-tempered scheme with a bias temperature of 3000 K. The simulation was run for 3.0 μs. The WTM-eABF simulation utilized three separate windows run in parallel for efficiency. The extended-Lagrangian formalism was set with a fluctuation amplitude of 0.1 Å and a bin width of 0.2 Å. The coupling time constants for the extended-Lagrangian for both CVs were set to be 200 fs. A minimum of 10,000 full samples per bin was required before integrating the mean force. The WT-MtD section had a hill width of 1.0 Å, hill weight of 0.1 kcal/mol, deposited every 2.0 ps and bias temperature of 3000 K. The total simulation time was 6.25 μs over all three windows. The long MW WTM-eABF reference was run for 8.0 μs. Last, the OPES simulations utilize Gaussian kernels of widths 1.0 Å and were deposited every 2.0 ps. Three simulations with different barrier settings of 10, 15 and 20 kcal/mol were run. For all three barrier settings, the widths were dynamically set based on the variance of the CV and a neighbor list was used for kernel summation. The PMF as a function of the *z*-projected distance was computed on the fly by reweighting. The simulation time run for each barrier setting was 3.0 μs.

nAChR channel permeation by sodium. The free-energy change underlying the translocation of a sodium ion within the pore of the human α₃β₄ nicotinic acetylcholine receptor was determined using as an RC model the Euclidean distance between the ion and the center of mass of the pore—computed from the trace of the five segments spanning residue 241 through 262—projected onto the longitudinal axis of the latter, i.e., the *z*-direction of Cartesian space. The reaction pathway spanned −22 to +22 Å, where *z* = 0 is the center of mass of the pore. In addition, the distance between the ion and center of mass of the pore orthogonal to the *z*-direction of Cartesian space was restrained to remain below 5.0 Å by applying an additional harmonic potential with a force constant of 10 kcal mol⁻¹ Å⁻². For the REUS^{17,82,83} simulations, initial conformations were generated as follows. Starting from the equilibrated complex at *z* = 0, two SMD⁷¹ simulations were performed in which the sodium ion was pulled along the channel axis to *z* = ±22 Å over 10-ns, with a force constant of 100 kcal mol⁻¹ Å⁻². We note that this SMD-based protocol was preferred over direct placement of the ion at successive *z*-positions because it generates starting configurations in which the local protein, water, and ion environment have partially relaxed along the permeation pathway, thereby reducing the equilibration time required within each umbrella window. From these trajectories, 88 configurations were extracted at *z* = 0.5 Å intervals and used as initial structures for subsequent one-dimensional REUS simulations. The reaction pathway was, thus, divided into 88 windows with centers spanning from −22 to +22 Å in 0.5-Å increments. Three copies were assigned to each window (88 × 3 replicas in total), and each replica was simulated for 5 ns, resulting in a cumulative simulation time of approximately 1.3 μs. A harmonic bias with a force constant of 12.5 kcal mol⁻¹ Å⁻² was applied in each window. Exchange attempts between neighboring windows were attempted every 4 ps, yielding an average exchange rate of approximately 10% across the whole reaction coordinate. The same non-parametric reweighting approach^{42,79}, as the one in the coupled protein-ligand binding/

conformational transition ribose-binding protein, was used to reconstruct the PMF. The WT-MtD simulations used Gaussian hills deposited every 4 ps, with a width of 0.1 Å and an initial height of 0.015 kcal/mol with a bias temperature of 2000 K. The PMF calculation was run for 2.0 μs and reweighted according to the procedure implemented in Colvars^{44,45}. For the WTM-eABF algorithm, the reaction pathway was discretized in bins 0.2 Å wide. The standard deviation between the CV and the fictitious particle was set to 0.1 Å, and the oscillation period of the fictitious particle to 100 fs. A threshold of 20,000 force samples per bin was defined prior to the application of the biasing force⁸⁴. The WT-MtD portion had a hill width of 1.0 Å, hill weight of 0.1 kcal/mol, deposited every 4.0 ps and bias temperature of 3000 K. The long MW WTM-eABF reference was run for 4.29 μs. The OPES simulations utilized Gaussian kernels with a width of 1.0 Å and were deposited every 4 ps. Two simulations with barrier settings of 5 and 10 kcal/mol were run for 2.0 μs each. For both barrier settings, the widths were dynamically set based on the variance of the CV and a neighbor list was used for kernel summation. The PMF as a function of the projected distance were computed on the fly with appropriate reweighting.

Error analysis and convergence assessment

Bootstrapping the error. The use of bootstrapping in this context is motivated by its interpretation as a resampling procedure that mimics the acquisition of new data from the same underlying distribution. Given the same system, all enhanced-sampling methods are expected, in the limit of sufficient sampling, to recover the same underlying FEL. Treating the set of method-specific FELs as independent samples drawn from this common distribution allows the bootstrap procedure to quantify the variability that would be expected upon repeating the calculations with alternative sampling strategies. The resulting uncertainty, thus, reflects a conservative estimate of the inter-method variability.

To estimate uncertainties in the reconstructed FELs, we employed a nonparametric bootstrap procedure. For each system, the final FEL obtained from method *m* ∈ {1, ..., *M*} (with *M* = 4 in this work) is denoted by *A_m*(ξ). All methods performed for a given system employ the same CV, ξ(**x**), and are placed on an identical discrete grid of *N* bins.

Each FEL was first converted into a normalized probability distribution,

$$P_m(\xi) = \frac{\exp(-\beta A_m(\xi))}{\sum_{j=1}^N \exp(-\beta A_m(\xi_j))}. \quad (14)$$

Bootstrap samples were generated by resampling, with replacement, *M* probability distributions from the set {*P_m*(ξ)}. For each bootstrap replica *b* ∈ {1, ..., *B*}, where *B* stands for the total number of resampled observables, a bootstrapped mean probability distribution was constructed as

$$\bar{P}^{(b)}(\xi) = \frac{1}{M} \sum_{m \in \mathcal{M}^{(b)}} P_m(\xi), \quad (15)$$

where $\mathcal{M}^{(b)}$ denotes the resampled set of methods in bootstrap replica *b*. The corresponding FEL was obtained as

$$\bar{A}^{(b)}(\xi) = -\frac{1}{\beta} \ln \bar{P}^{(b)}(\xi) + A_0, \quad (16)$$

which is defined up to an additive constant *A*₀.

The final reported FEL, referred to as the marginalized average, was taken as the mean over all bootstrap replicas,

$$\langle A(\xi) \rangle = \frac{1}{B} \sum_{b=1}^B \bar{A}^{(b)}(\xi), \quad (17)$$

and the associated uncertainty as the standard deviation across the bootstrap ensemble,

$$\sigma_A(\xi) = \left[\frac{1}{B-1} \sum_{b=1}^B \left(\bar{A}^{(b)}(\xi) - \langle A(\xi) \rangle \right)^2 \right]^{1/2} \quad (18)$$

Convergence assessments. For each enhanced-sampling method, convergence was evaluated using two complementary criteria, (1) a self-convergence metric and comparison to the corresponding marginalized average, and (2) independent long WTM-eABF reference simulations. Self-convergence was quantified using the RMSD of the FEL relative to its final estimate. Let $A(\xi, t)$ denote the FEL evaluated at simulation time t as a function of CV $\xi(\mathbf{x})$, and $A(\xi, t_\infty)$ the final FEL obtained from the completed trajectory. The FEL was evaluated on a fixed discretization of the CV space into N bins, indexed by ξ_i . Since the free energy is defined up to an arbitrary additive constant, each FEL was vertically aligned to the final surface by subtracting the mean difference between the two surfaces, yielding

$$\tilde{A}(\xi_i, t) = A(\xi_i, t) - \frac{1}{N} \sum_{j=1}^N [A(\xi_j, t) - A(\xi_j, t_\infty)] \quad (19)$$

minimizing the squared difference between $A(\xi, t)$ and $A(\xi, t_\infty)$, ensuring maximal overlap between the two FELs. The self-convergence at time t is then defined as

$$\text{RMSD}_{\text{self}}(t) = \left[\frac{1}{N} \sum_{i=1}^N \left(\tilde{A}(\xi_i, t) - A(\xi_i, t_\infty) \right)^2 \right]^{1/2} \quad (20)$$

and was monitored as a function of time. A simulation was considered self-converged when $\text{RMSD}_{\text{self}}(t)$ reached a stable plateau $\leq k_B T$.

In addition to the self-convergence criterion, we also assessed convergence with respect to the marginalized average FEL. As was done for the self-convergence analysis, the instantaneous FEL was first vertically aligned to the reference surface by subtracting the mean difference between the two surfaces, ensuring maximal overlap. The RMSD relative to the marginalized average was then computed as

$$\text{RMSD}_{\text{ref}}(t) = \left[\frac{1}{N} \sum_{i=1}^N \left(\tilde{A}(\xi_i, t) - \langle A(\xi_i) \rangle \right)^2 \right]^{1/2} \quad (21)$$

While self-convergence quantifies the internal stability of a given enhanced-sampling method, convergence relative to the marginalized average provides a measure of consistency with the consensus FEL recovered across all methods.

The second reference consists of independent, long MW WTM-eABF simulations (multi- μ s), which were performed for each system. These extended trajectories were not used in the construction of the marginalized averages, but, instead, served to assess whether the consensus profiles recovered the same thermodynamic features observed under prolonged sampling. Agreement between the marginalized averages and the long WTM-eABF references was also quantified using an RMSD metric.

It is important to note that the consensus FEL construction described above does not anchor profiles at any single state. By converting each method's FEL into a normalized probability distribution, the arbitrary additive constant is absorbed through normalization, and the bootstrap procedure propagates uncertainties uniformly across the full CV range. An alternative approach of anchoring all profiles at a common reference state in free-energy space would force the

uncertainty to zero at the anchor point by construction. This results in a concentration of errors into regions farther from this reference point, producing an error distribution that reflects the choice of anchor rather than the true inter-method variability.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Initial structures used in this study are available in the Protein Data Bank: 1BBZ [10.2210/pdb1BBZ/pdb] for the Abl-SH3:p41 complex; 1DRJ [10.2210/pdb1DRJ/pdb] for the ribose-binding protein; 5MO4 [10.2210/pdb5MO4/pdb] and 1OPL [10.2210/pdb1OPL/pdb] for BCR-Abl1 kinase; and 6PV7 [10.2210/pdb6PV7/pdb] for the $\alpha_3\beta_4$ nicotinic acetylcholine receptor. The input files supporting this study have been deposited at Zenodo [10.5281/zenodo.18599380] and are also available upon request. The deposit covers all five biomolecular processes and all four enhanced-sampling schemes investigated here, and includes topology files (PSF), Protein Data Bank (PDB) files, starting configuration files (Cartesian coordinates, velocities, and box dimensions), NAMD configuration files, Colvars configuration files, and force-field parameter files. For REUS simulations, starting configuration files for all windows are additionally provided. Additional data, including convergence analyses for each biomolecular process and enhanced-sampling scheme and the multi- μ s multiple-walker WTM-eABF reference simulations, are provided in the Supplementary Information. Source data are provided with this paper.

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Author contributions

C.C. conceived the study, C.C. and C.K. wrote the manuscript. H.C. implemented OPES in the Colvars library. J.C.G., M.M., C.C., and H.F. supervised the REUS, WT-MtD, WTM-eABF, and OPES calculations, respectively. H.F. designed the Abl-SH3:p41 binding assay; M.M. designed the ribose-binding protein assay; J.C.G. designed the BCR-Abl1 kinase DFG-flip assay; and C.C. designed the 2', 3'-dideoxyadenosine membrane permeation and nAChR sodium permeation assays. Individual calculations were performed as follows. Abl-SH3:p41: L.Y. (REUS), A.F. (WT-MtD), S.C.A. (WTM-eABF), and M.Z. (OPES); ribose-binding protein: A.F. (REUS), C.K. (WT-MtD), C.T. (WTM-eABF), and S.C.A. (OPES); BCR-Abl1 kinase: L.Y. (REUS), C.G.C. (WT-MtD), C.C. (WTM-eABF), and C.K. (OPES); 2', 3'-dideoxyadenosine membrane permeation: J.H. (REUS and WTM-eABF), C.T. (WT-MtD), and C.K. (OPES); nAChR sodium permeation: A.F. (REUS), C.K. (WT-MtD and OPES), and C.C. (WTM-eABF). R.S. and B.R. provided project oversight. All authors discussed the results and reviewed the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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